EE477: Project Report

16-bit Multiply-and-Accumulate (MAC) Circuit

by
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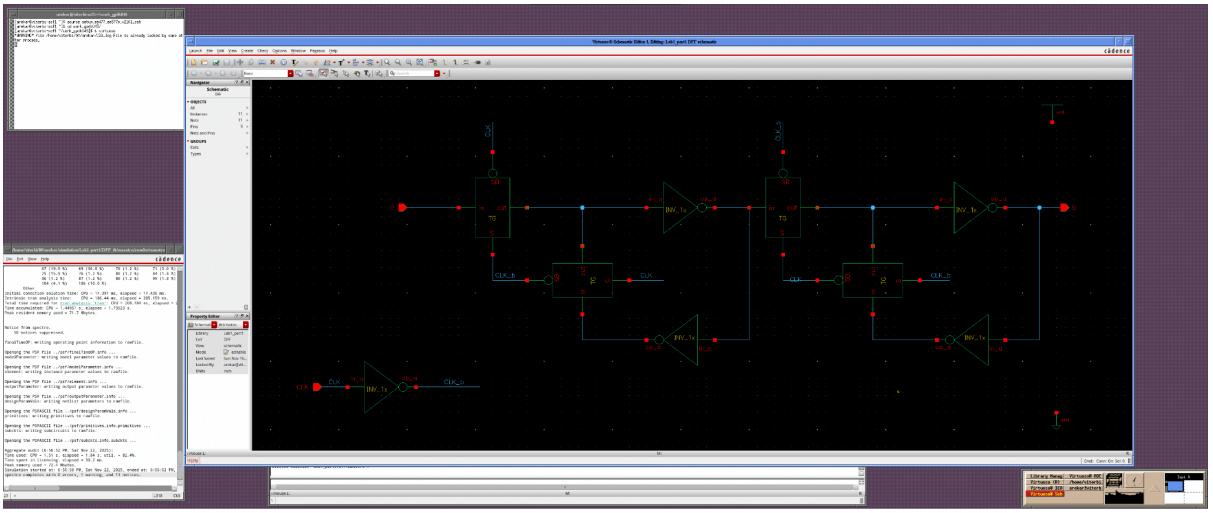
Smit Arekar
Vejaysuriyaa S

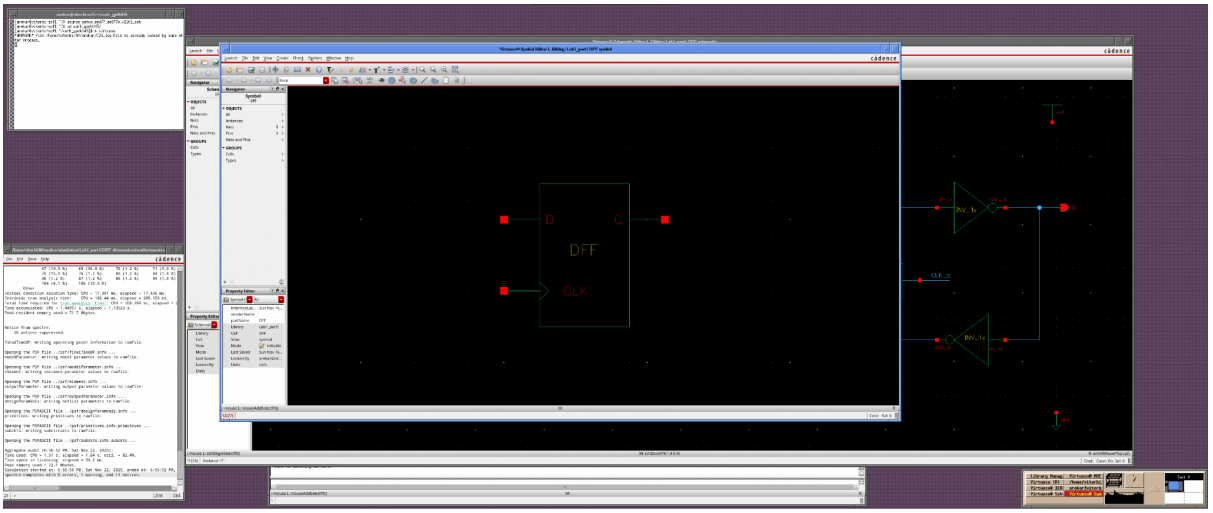
PHASE - I

Part A: DFF without reset

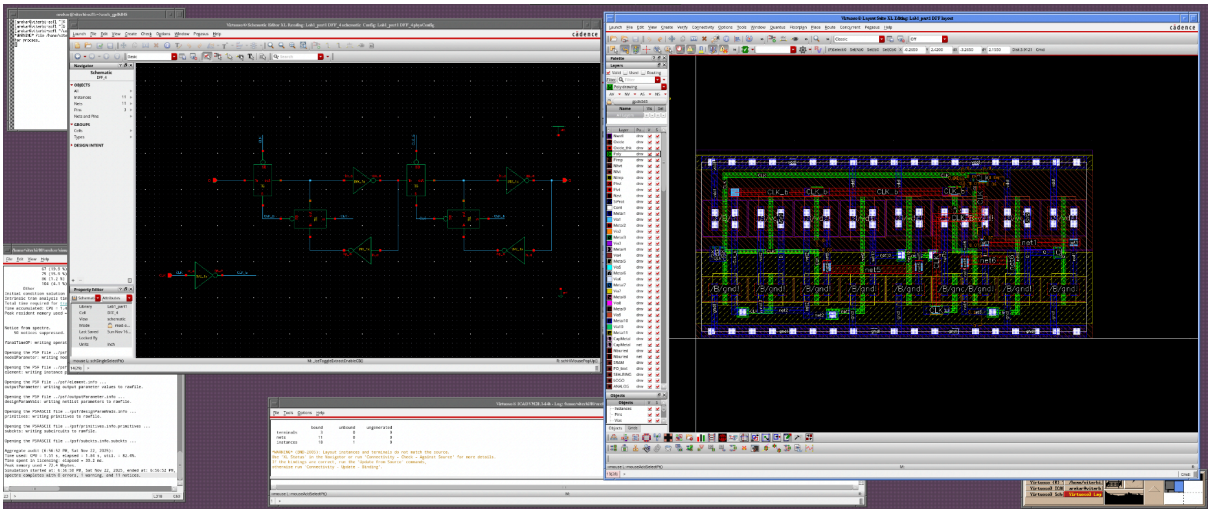
Schematic



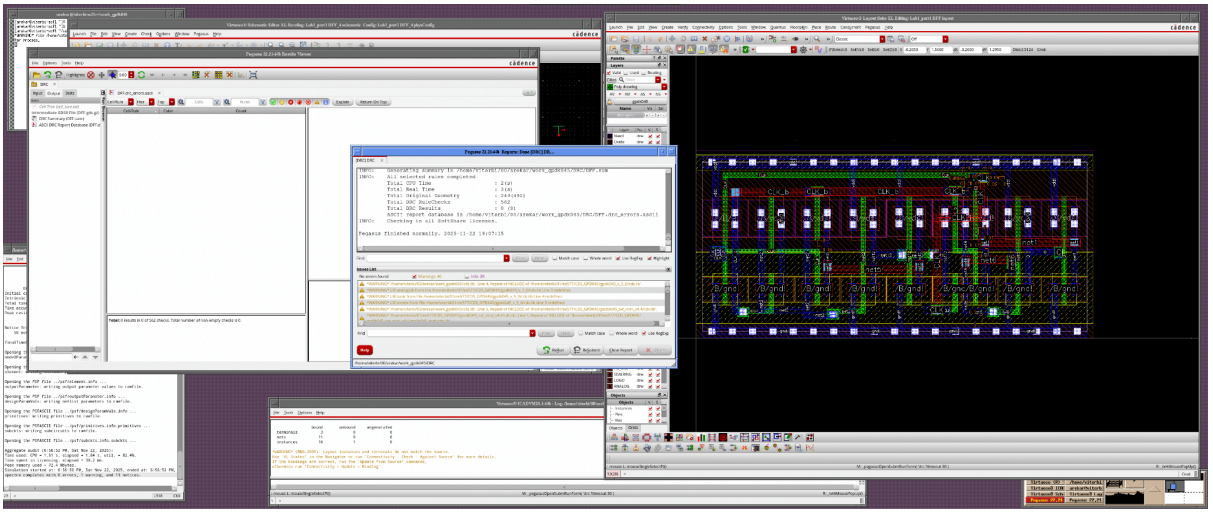
Symbol



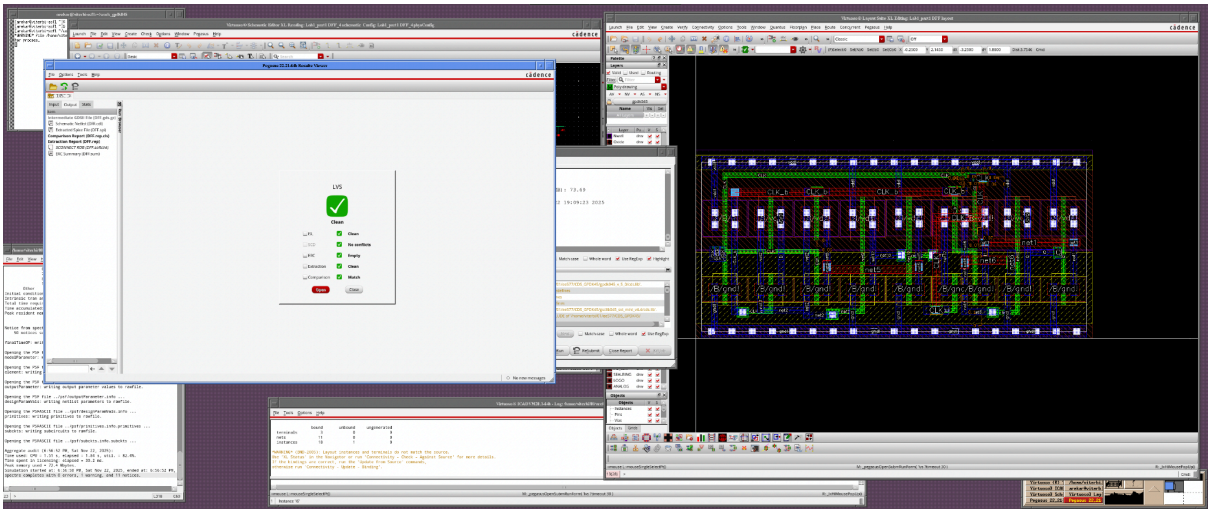
Layout



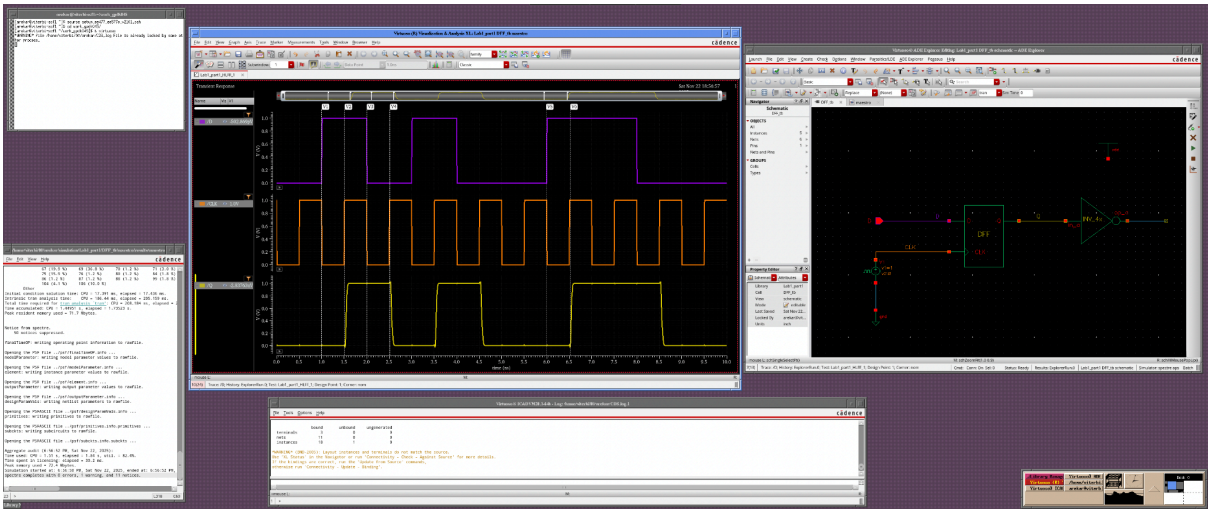
DRC Report



LVS reports



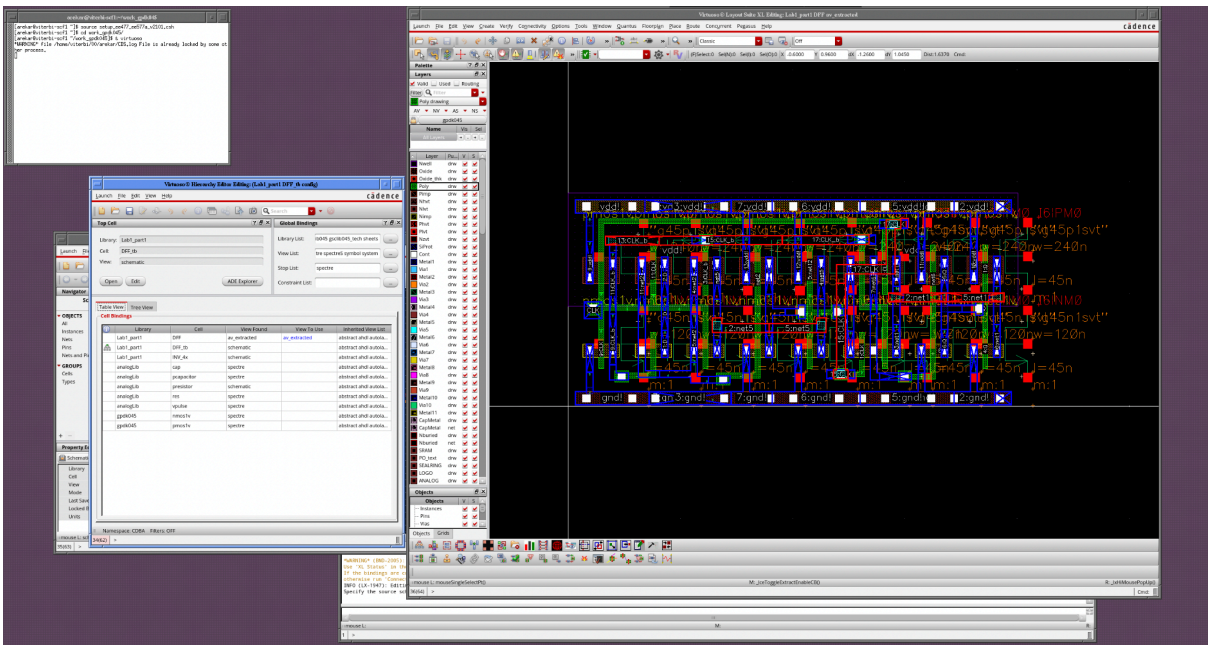
Functionality verification



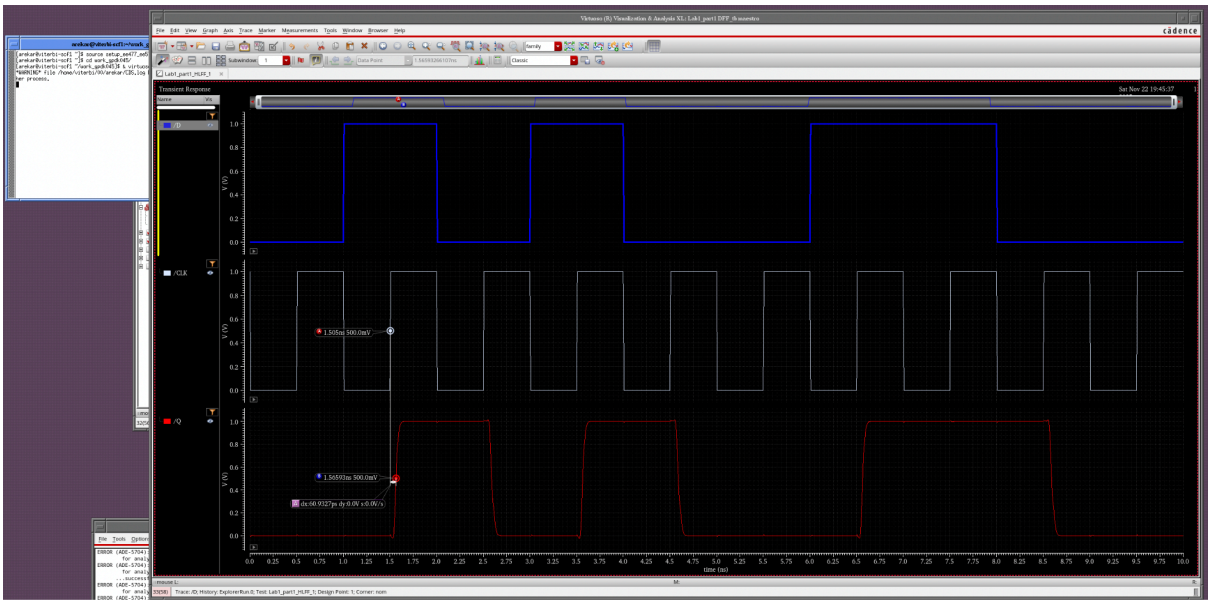
Cycle n	Time window	D_n (held stable)	Q_n (current)	Q_{n+1} (after next rising edge)	Transition check
0	0 ns → 1 ns	0	0	0	init/settle
1	1 ns → 2 ns	1	0	1	$(D_1,Q_1)=(1,0) \rightarrow (0,1)$
2	2 ns → 3 ns	0	1	0	$(D_2,Q_2)=(0,1) \rightarrow (1,0)$
3	3 ns → 4 ns	1	0	1	Q goes high after rise
4	4 ns → 5 ns	0	1	0	Q returns low
5	5 ns → 6 ns	0	0	0	stays low
6	6 ns → 7 ns	1	0	1	Q goes high
7	7 ns → 8 ns	1	1	1	stays high
8	8 ns → 9 ns	0	1	0	Q goes low

Delay Measurement

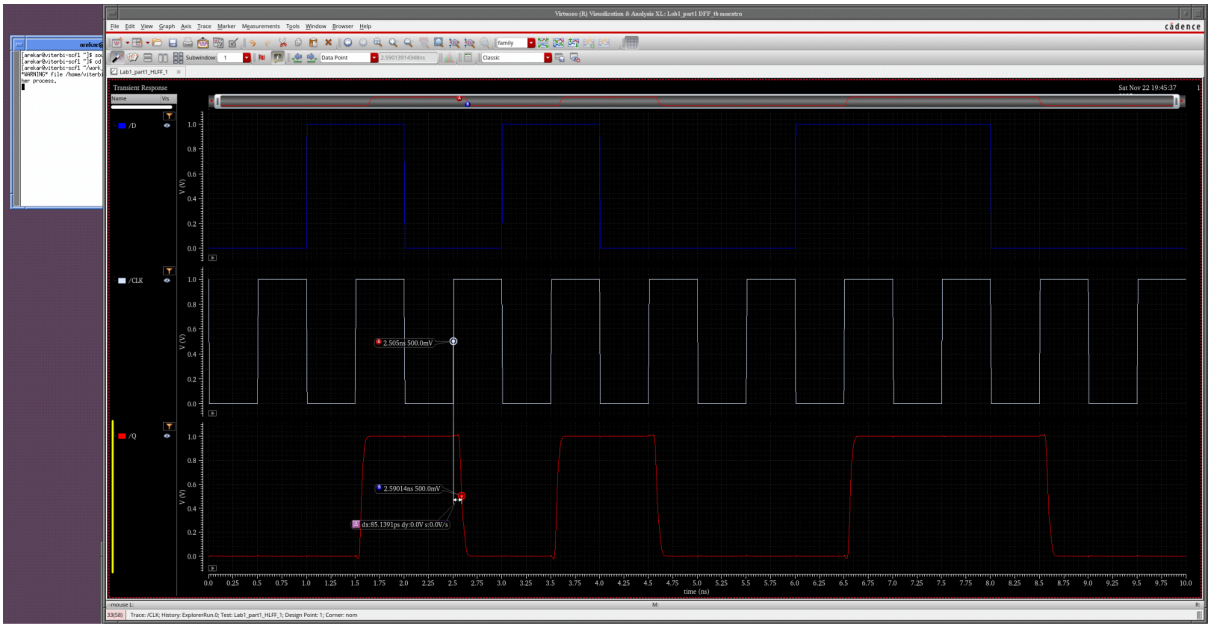
Extracted View



$(CLK_2,Q_1) = (0,0) \rightarrow (1,1)$



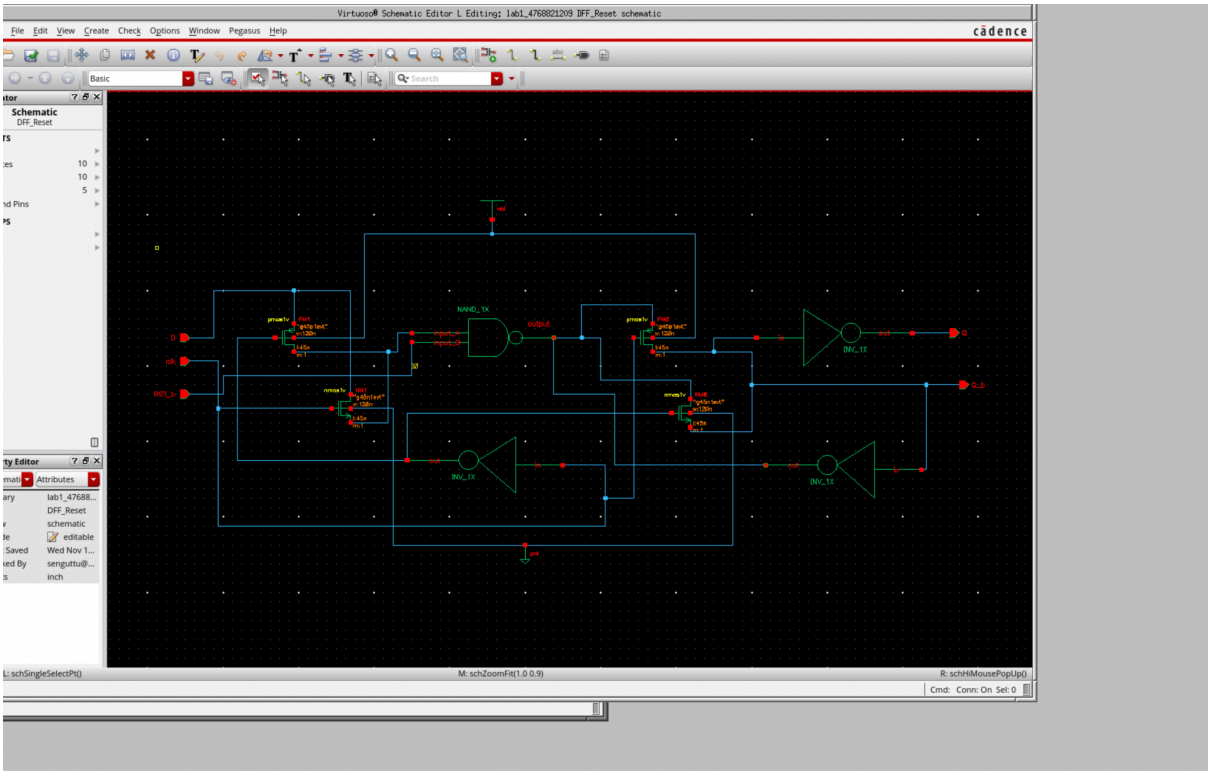
$(CLK_3,Q_1) = (0,1) \rightarrow (1,0)$



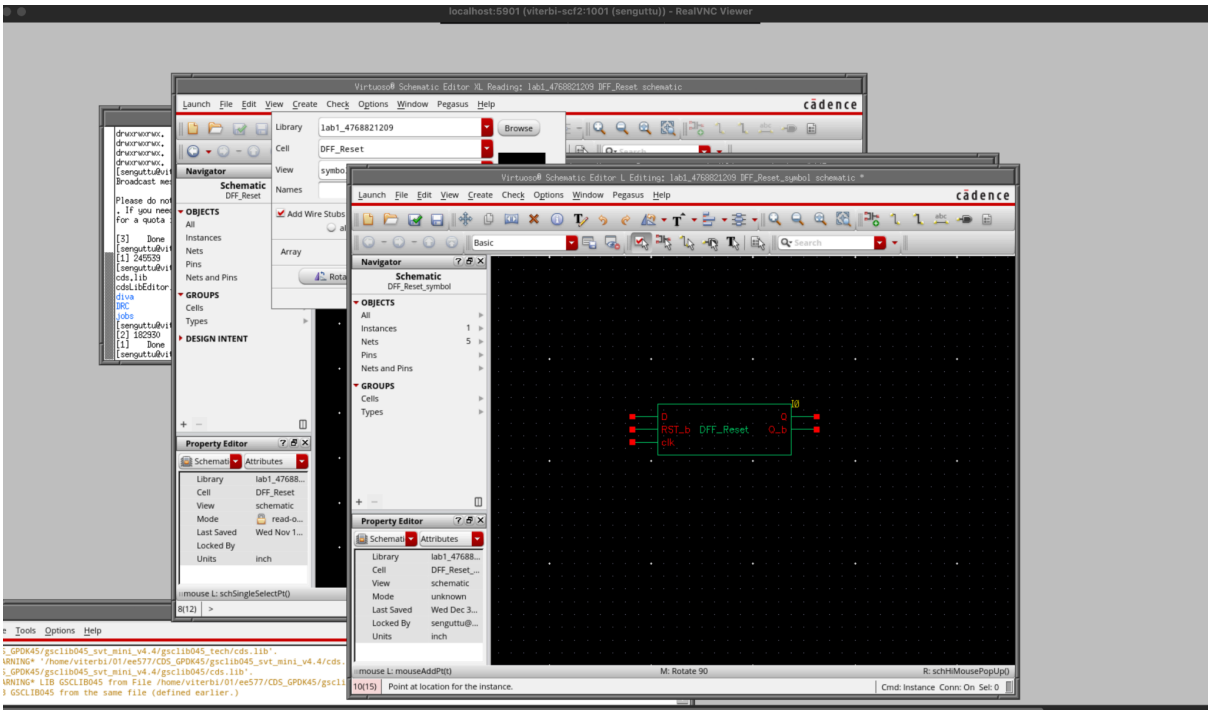
Time window	CLK_n	Q_n (current)	Q_{n+1} (after next rising edge)	Transition	Delay (ps)
1 ns → 2 ns	2	0	1	$(CLK_2,Q_1)=(0,0) \rightarrow (1,1)$	60.9327 ps
2 ns → 3 ns	3	1	0	$(CLK_3,Q_1)=(0,1) \rightarrow (1,0)$	85.1391 ps

Part A: DFF with Reset

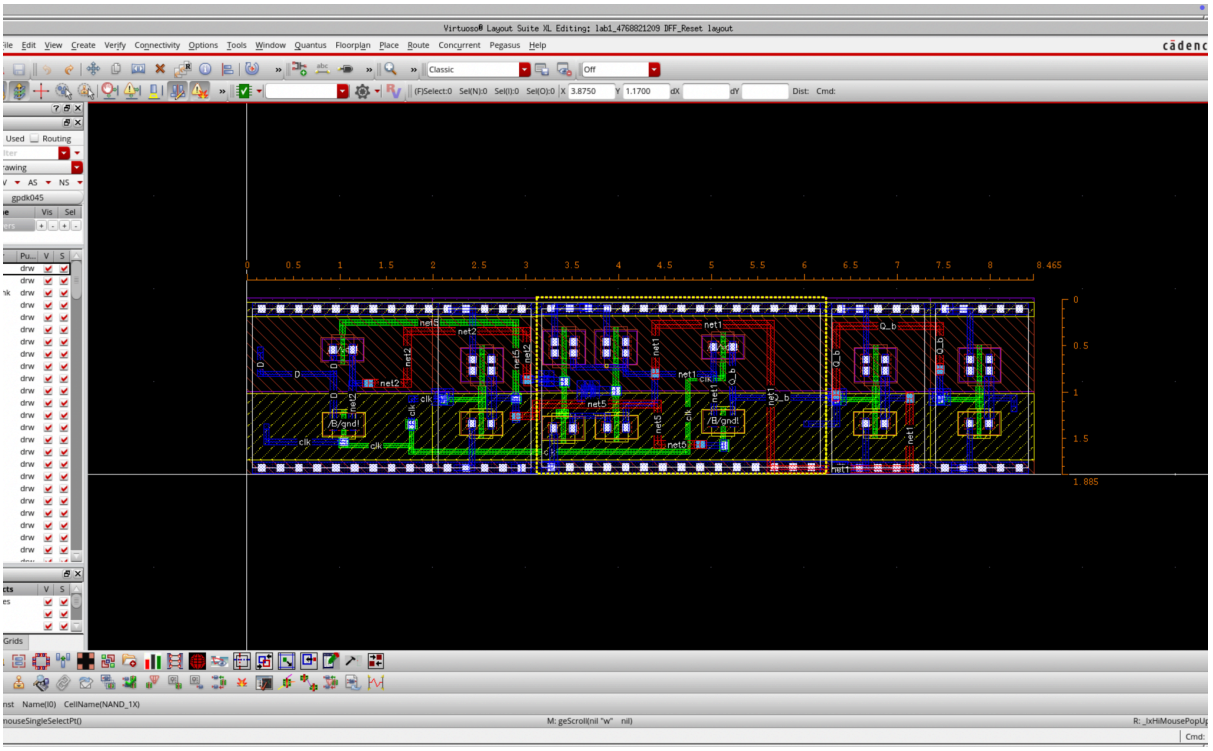
Schematic



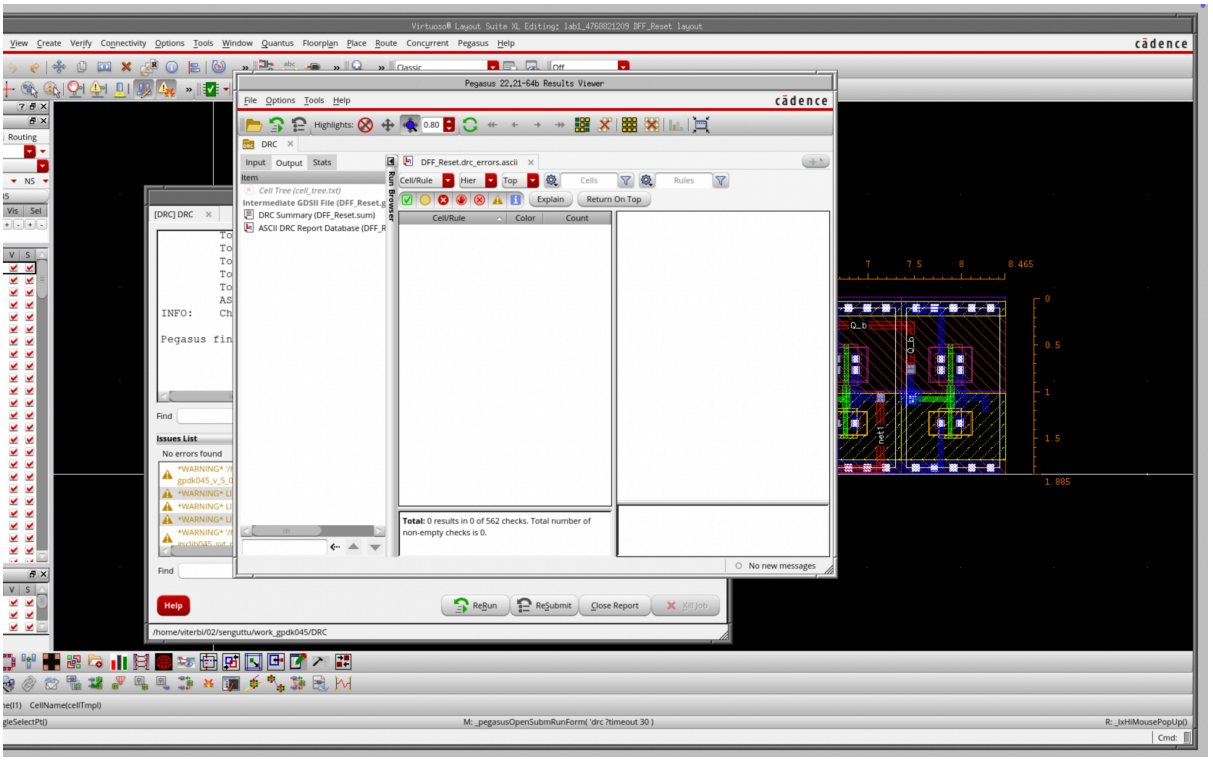
Symbol



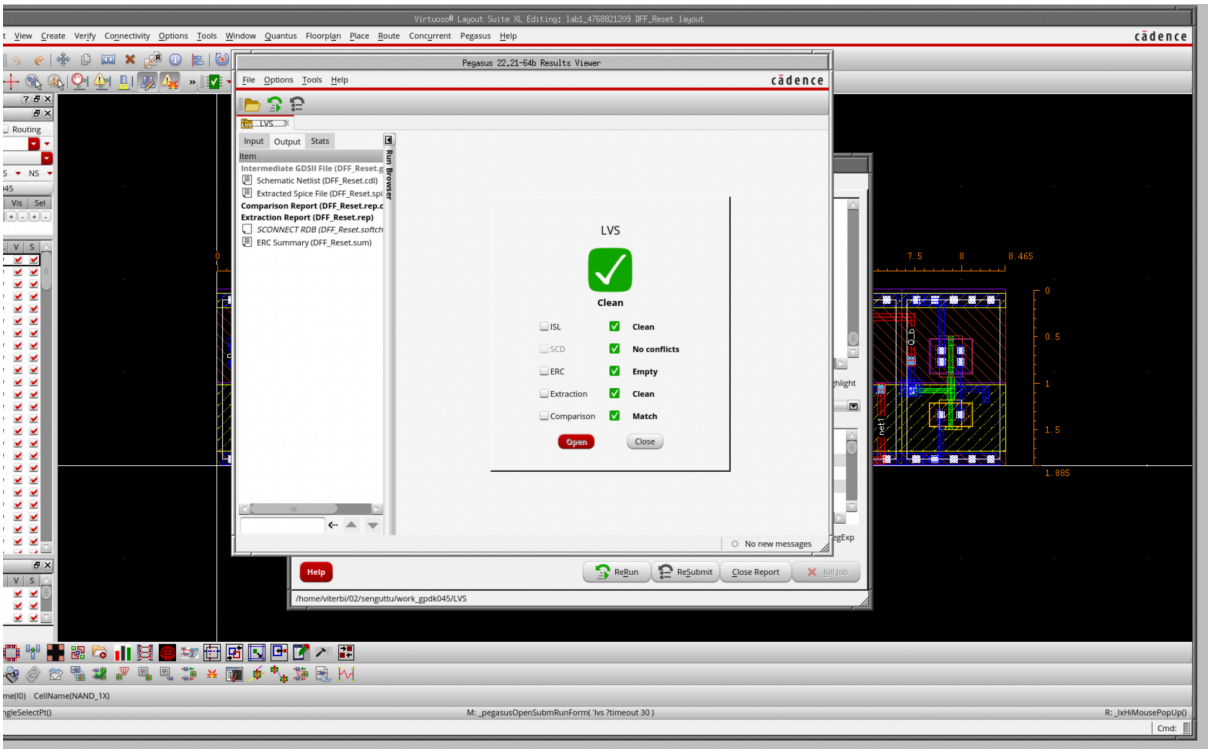
Layout



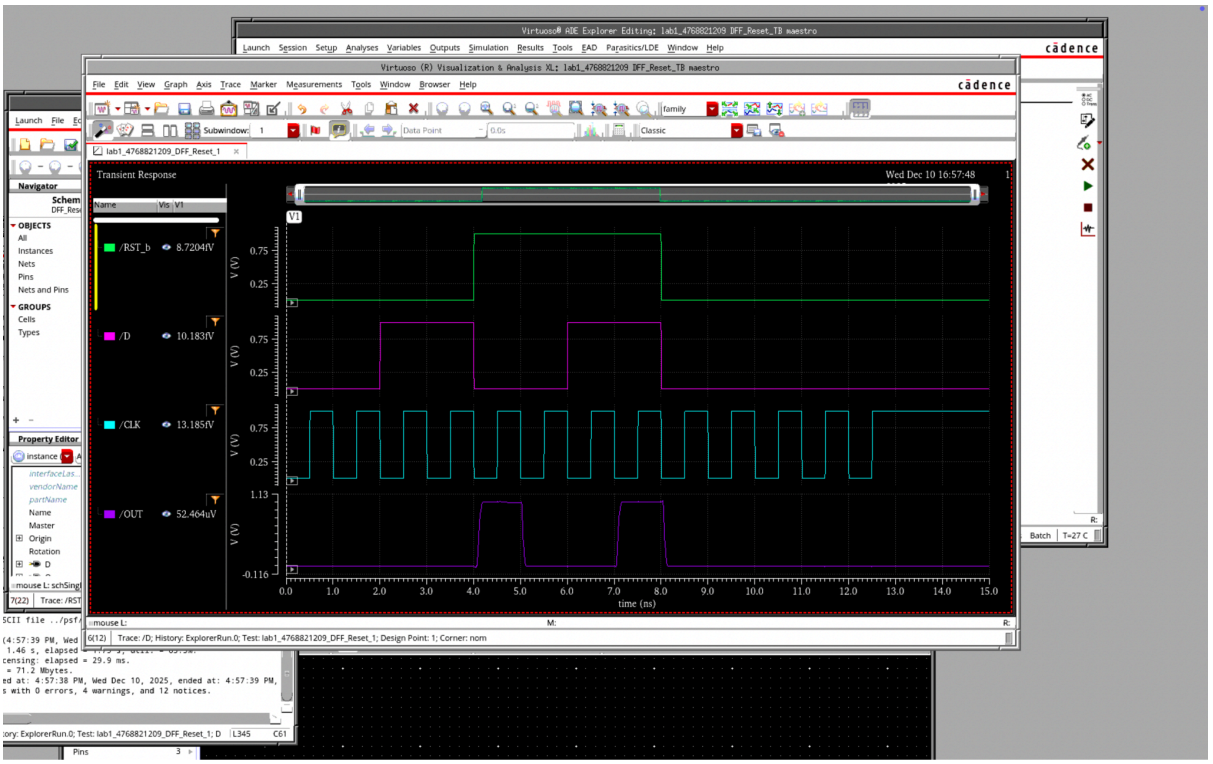
DRC Report



LVS Report



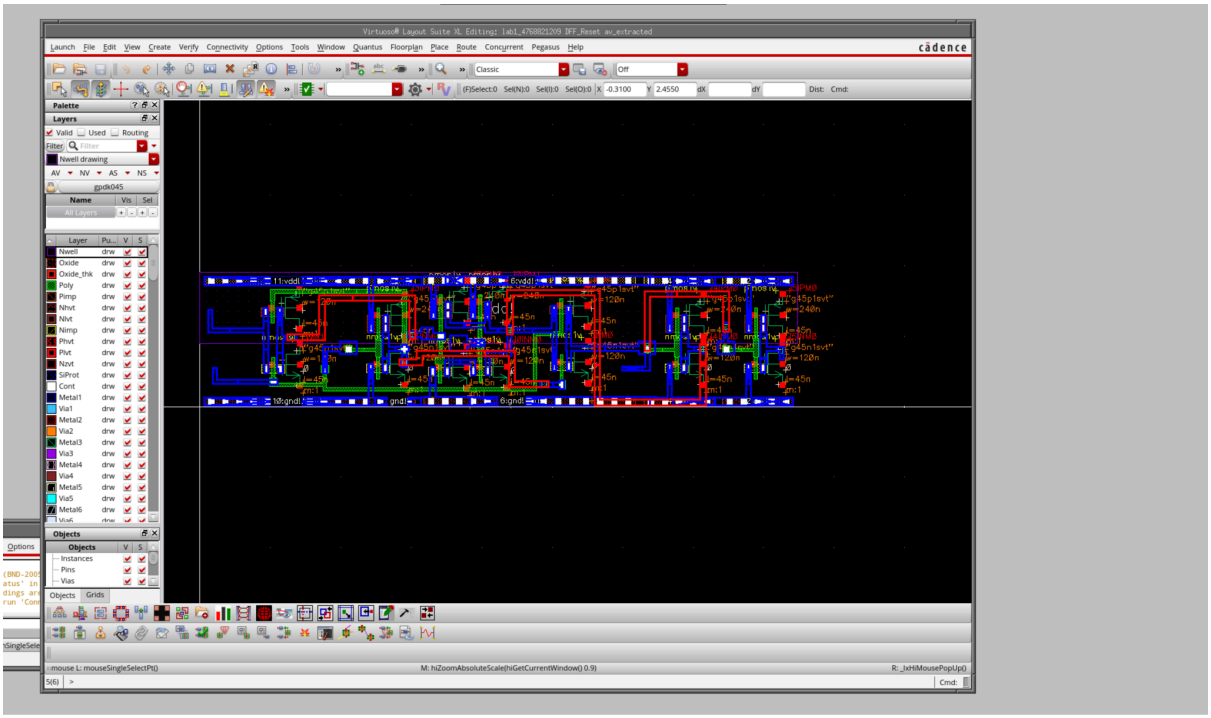
Functionality Verification



Cycle n	Time window (ns)	RST_b	D_n (held stable)	Q_n (current)	Q_{n+1} (after next rising edge)	Transition check
0	0 → 0.5	0	0	0	0	Reset active, Q f…
1	0.5 → 1.5	1	0	0	0	Q stays low (D=0)
2	1.5 → 2.5	1	0	0	0	Q stays low (D=0)
3	2.5 → 3.5	1	0	0	0	Q stays low (D=0)
4	3.5 → 4.5	1	1	0	1	(D,Q□) = (1,0) …
5	4.5 → 5.5	1	1	1	1	Q stays high (D=1)
6	5.5 → 6.5	1	0	1	0	(D,Q□) = (0,1) …
7	6.5 → 7.5	1	0	0	0	Q stays low (D=0)
8	7.5 → 8.5	0	0	0	0	Reset active, Q f…

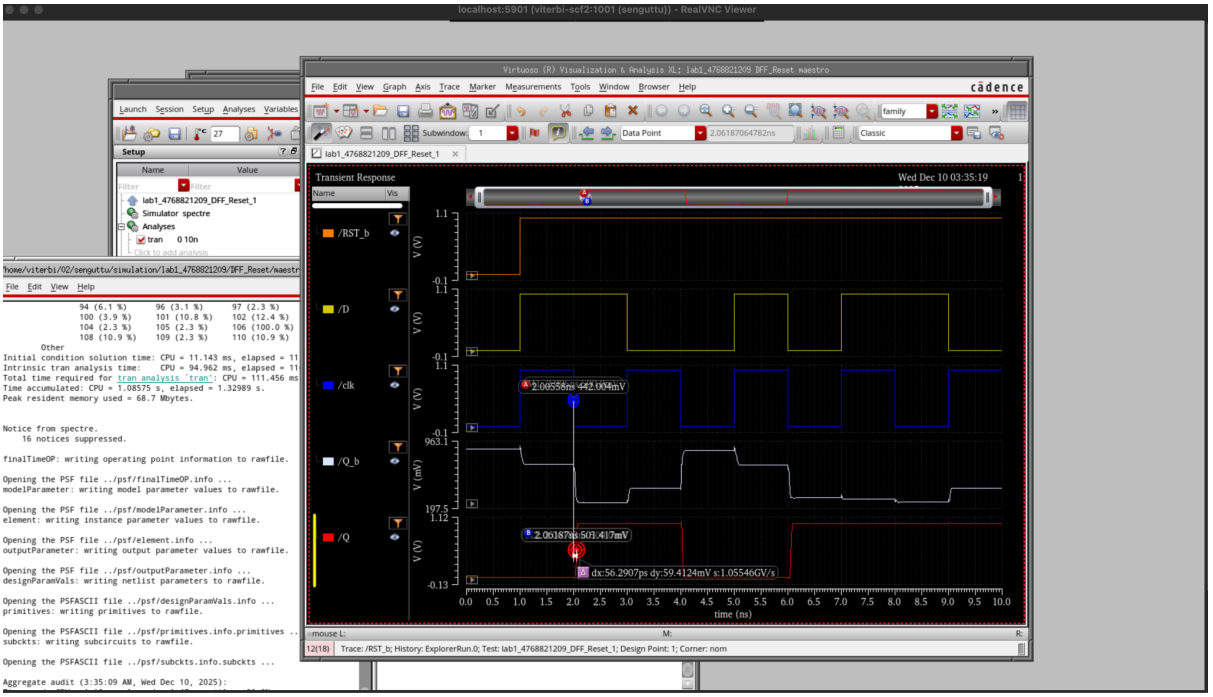
Delay Measurement

Extracted View

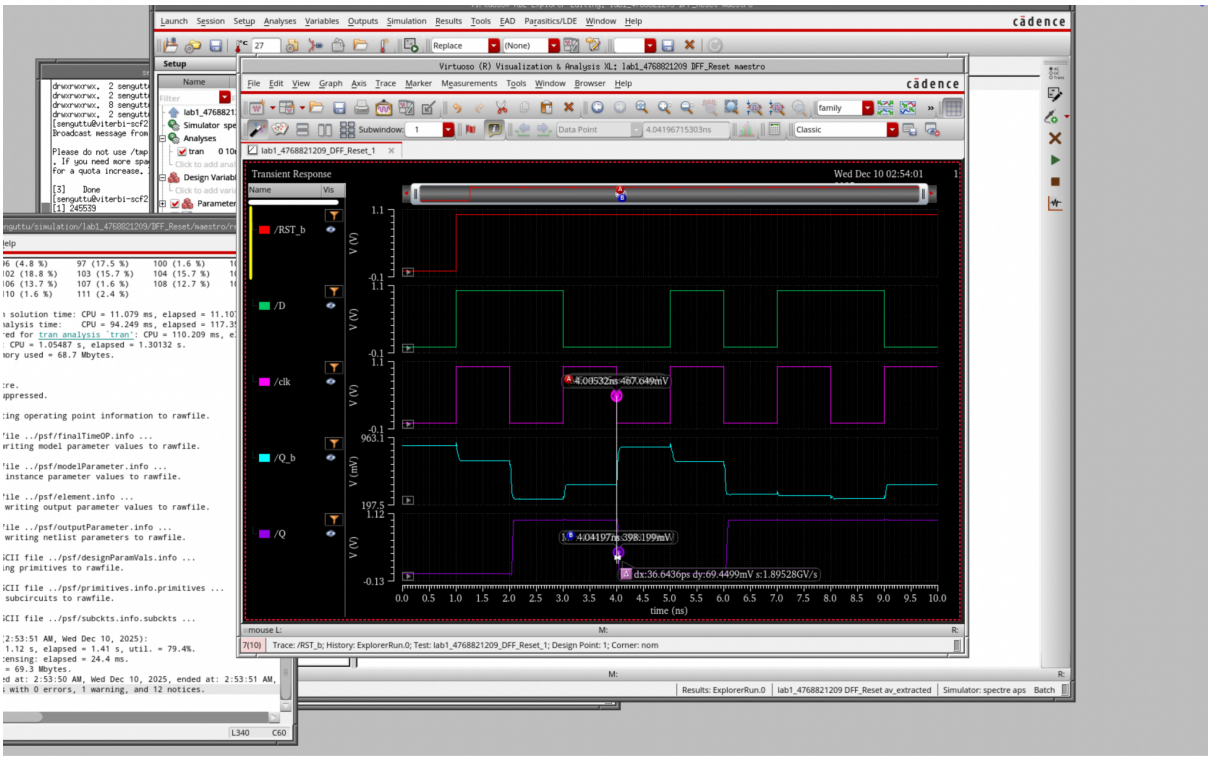


Measurement Type	Time of CLK Edge (ns)	Time of Q Transition (ns)	Transition	Measured Δt (ps)	Notes
Rising Delay (CLK → Q)	2.00528 ns	2.06179 ns	Q: 0 → 1	56.29 ps	From 50% CLK rising to 50% Q rising
Falling Delay (CLK → Q)	4.00532 ns	4.04197 ns	Q: 1 → 0	36.64 ps	From 50% CLK rising to 50% Q falling

Waveform for (CLK_2, Q_1): (0,0) → (1,1)

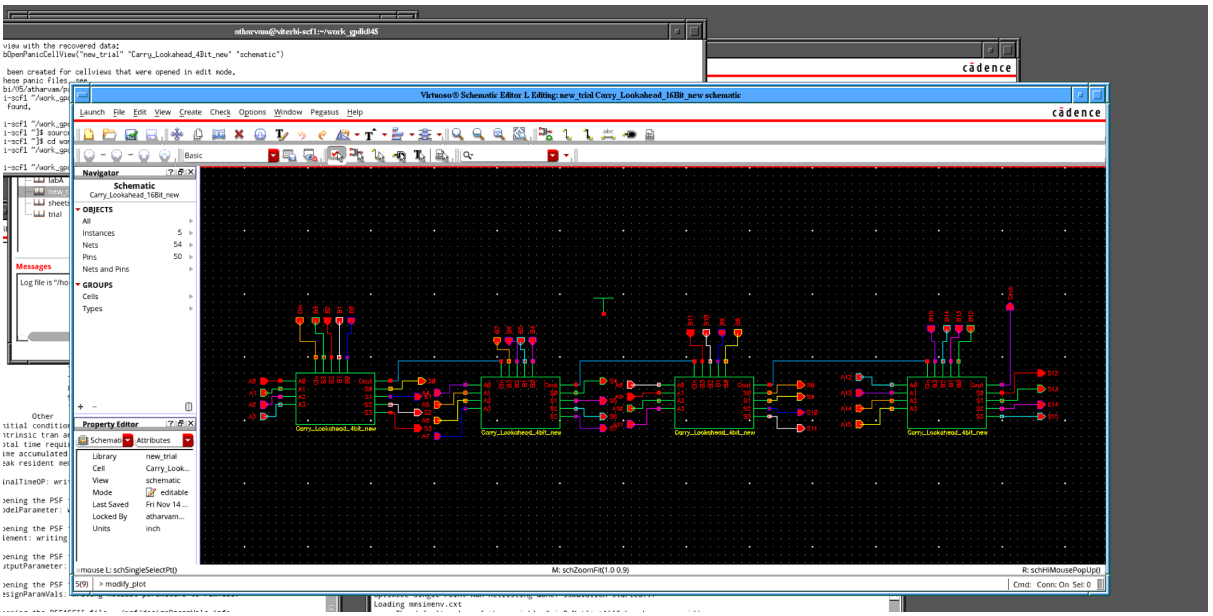


Waveform for (CLK_3, Q_1): (0,1) → (1,0)

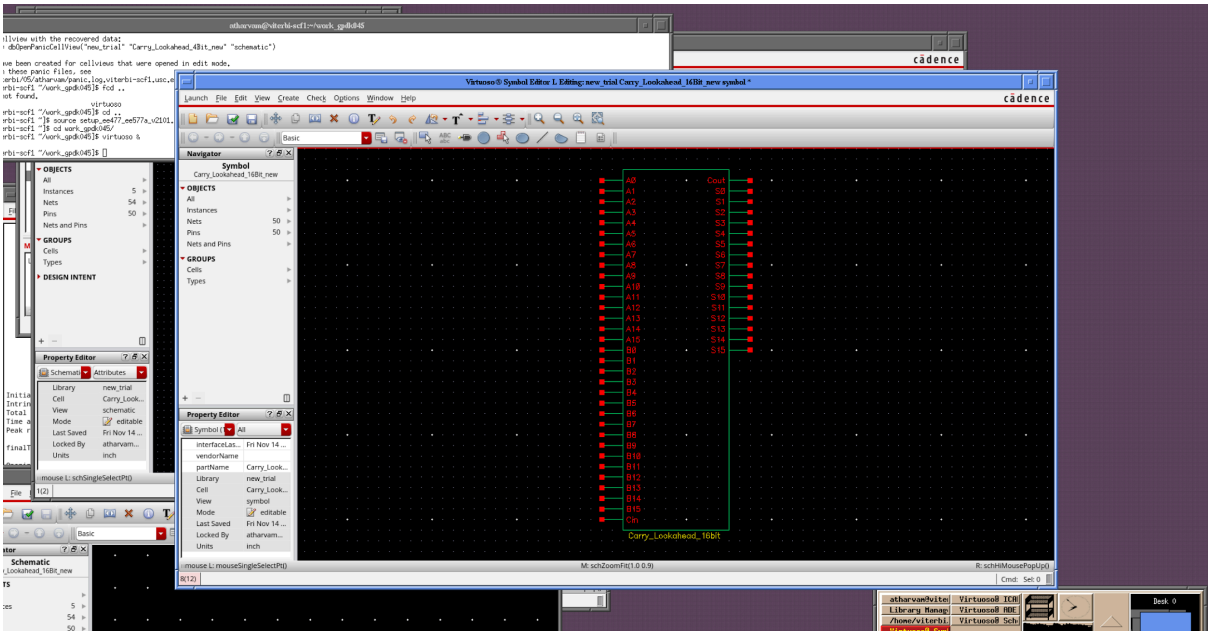


Part B: 16-bit adder

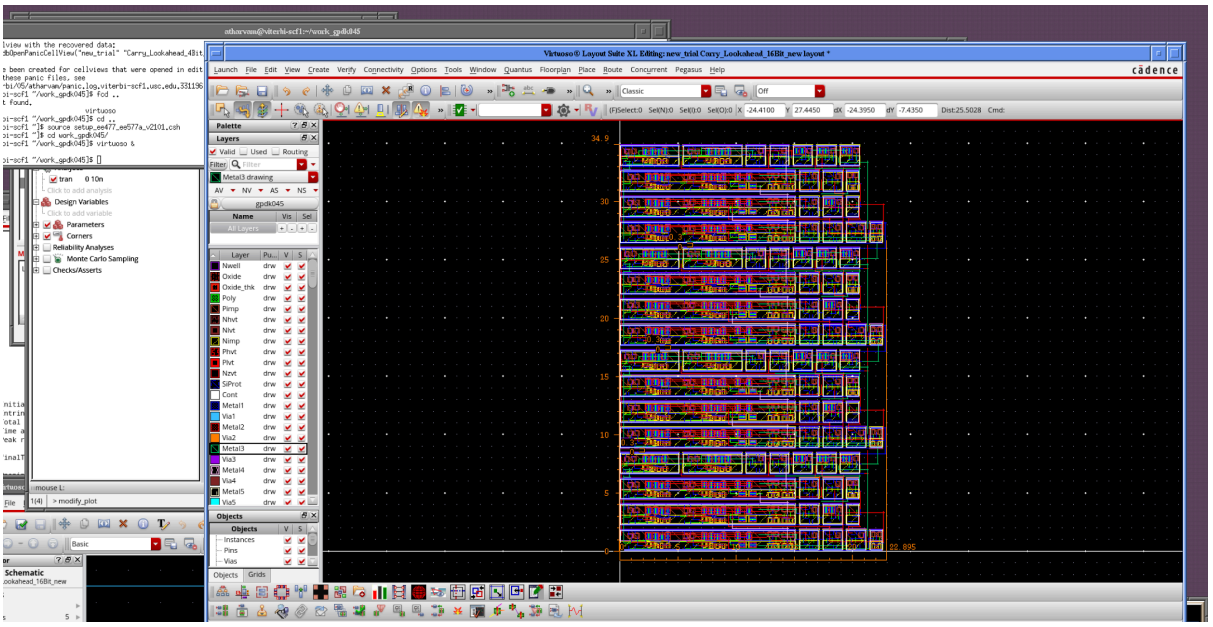
Schematic



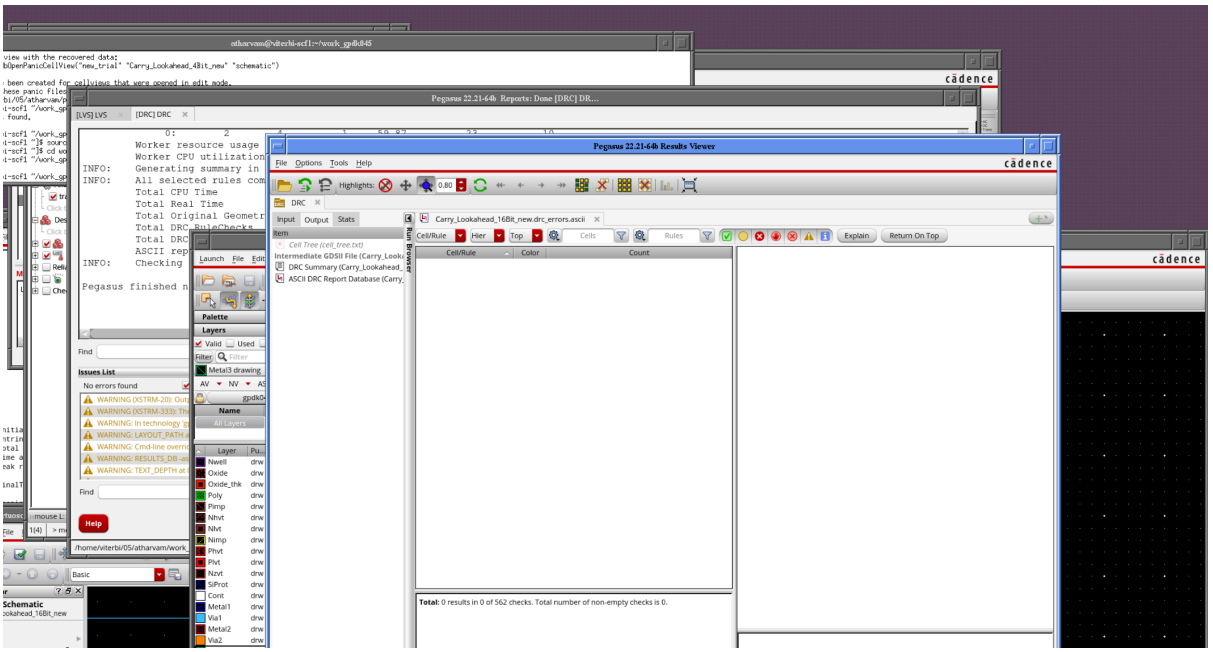
Symbol



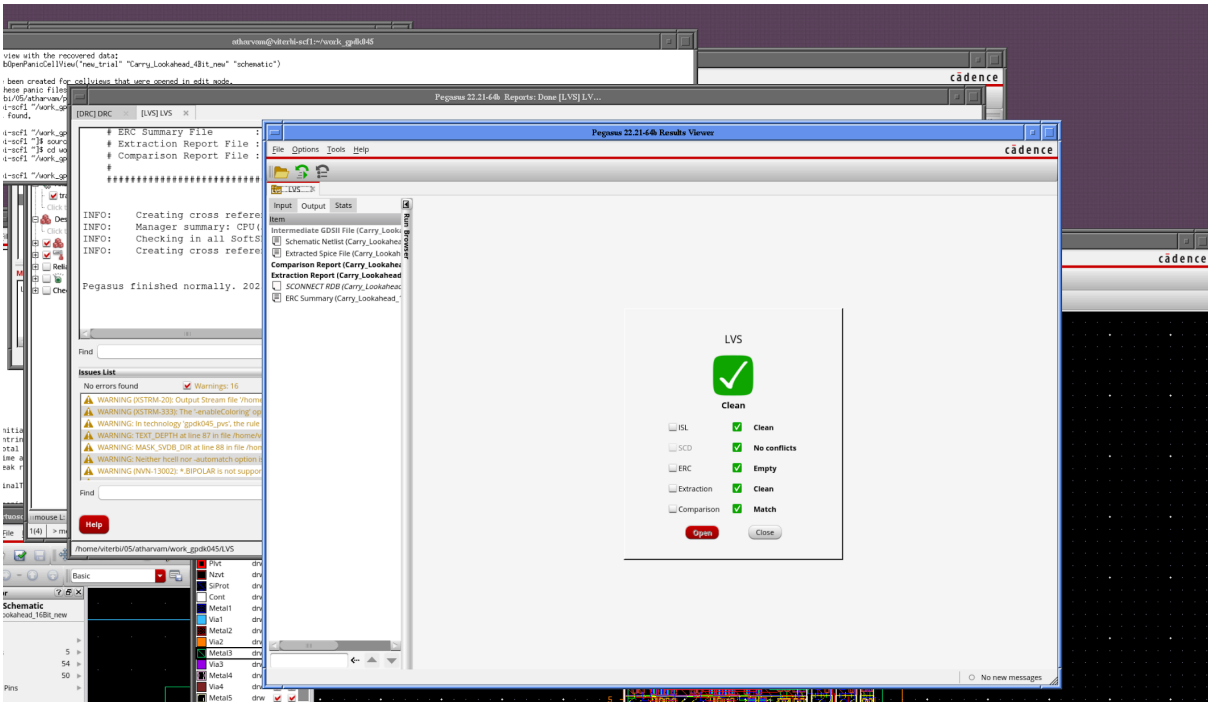
Layout



DRC Report



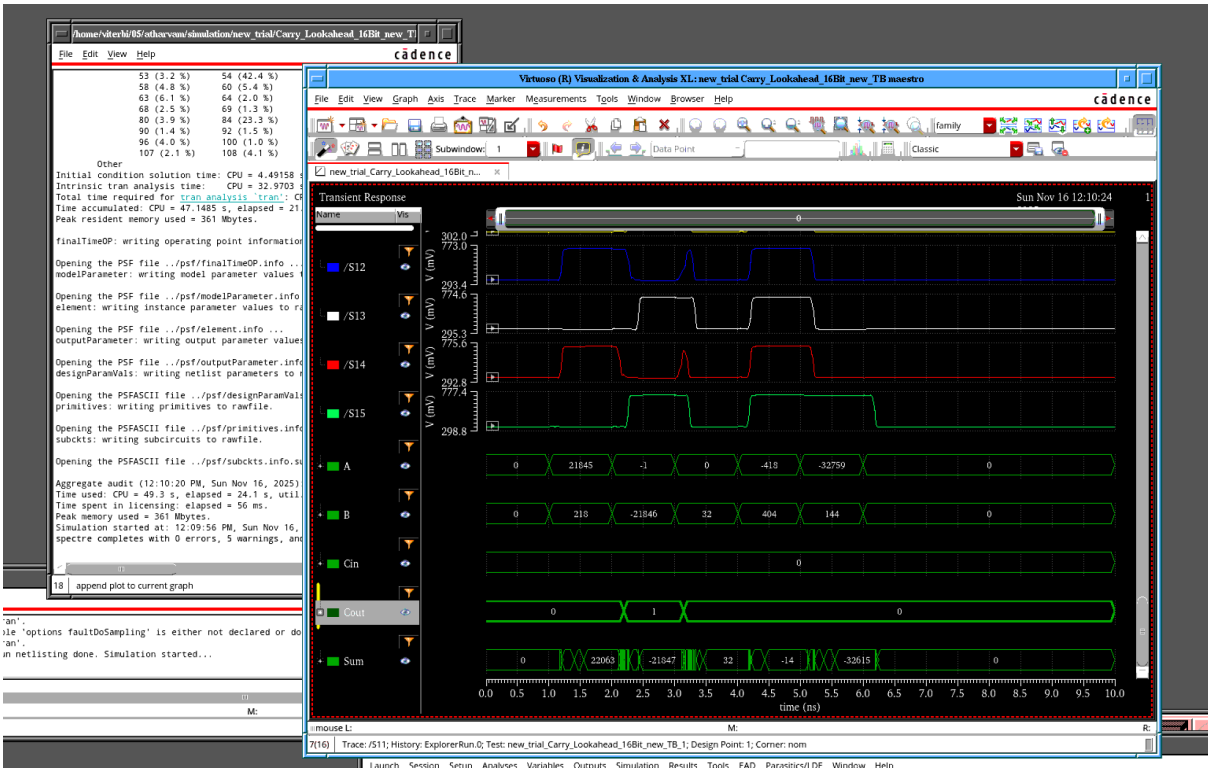
LVS Report



Functionality Verification

A. Schematic Testing

B. Post Layout Simulation



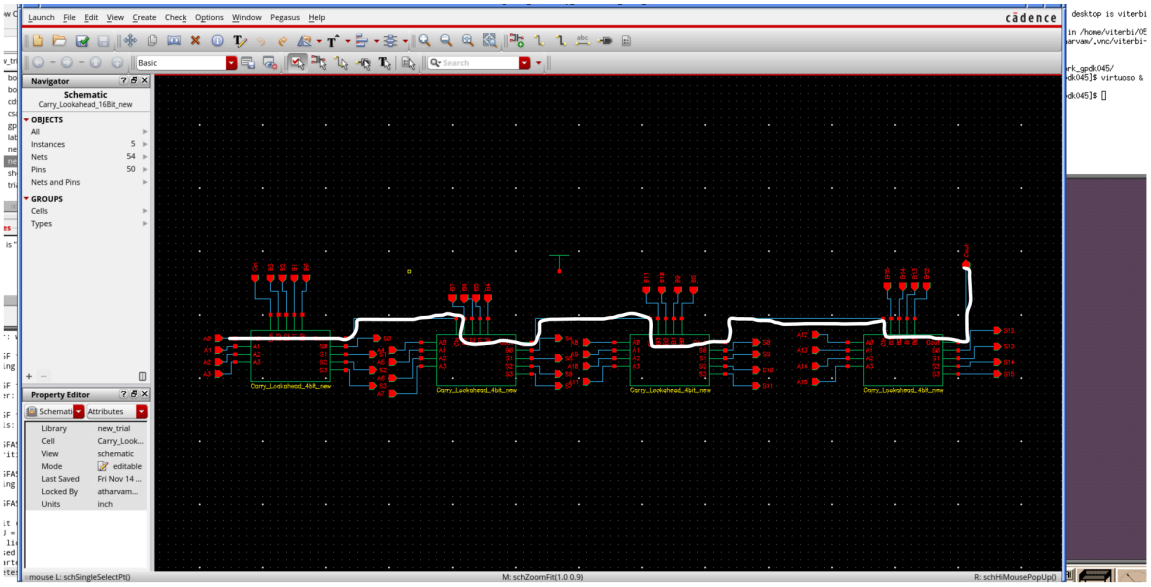
Critical Path → A0 to Cout

Critical Path – highlighted in white

Number of gates for critical path

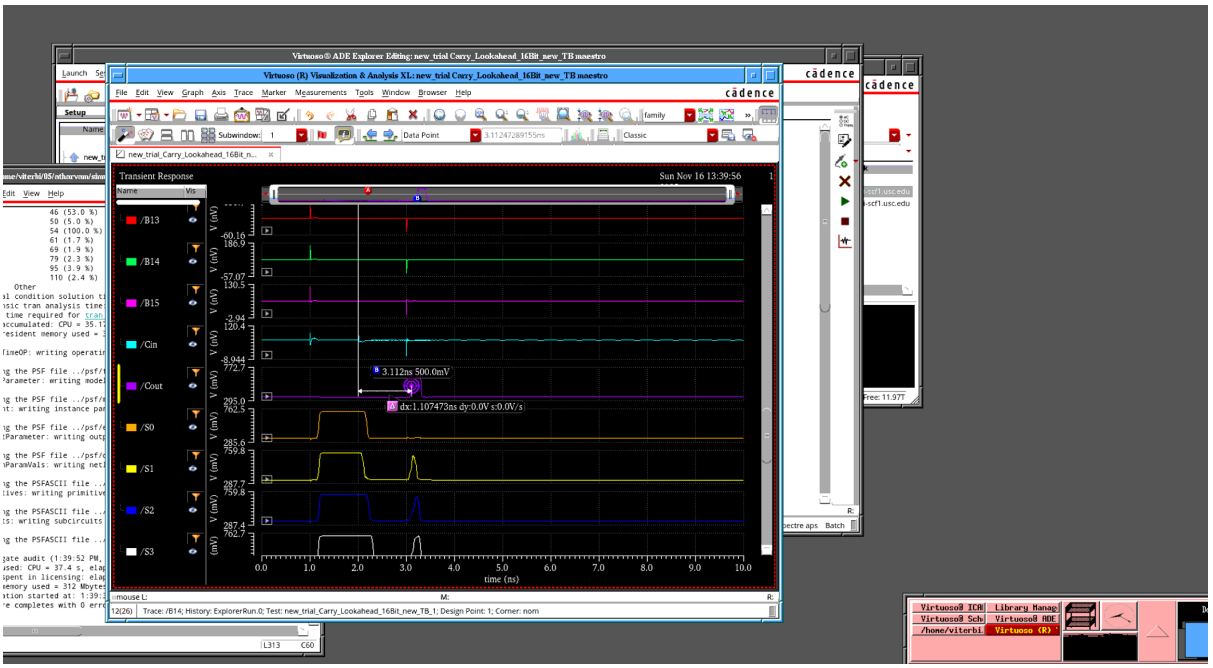
8 AND gates
7 OR gates
1 NAND gate
1 EXOR gate

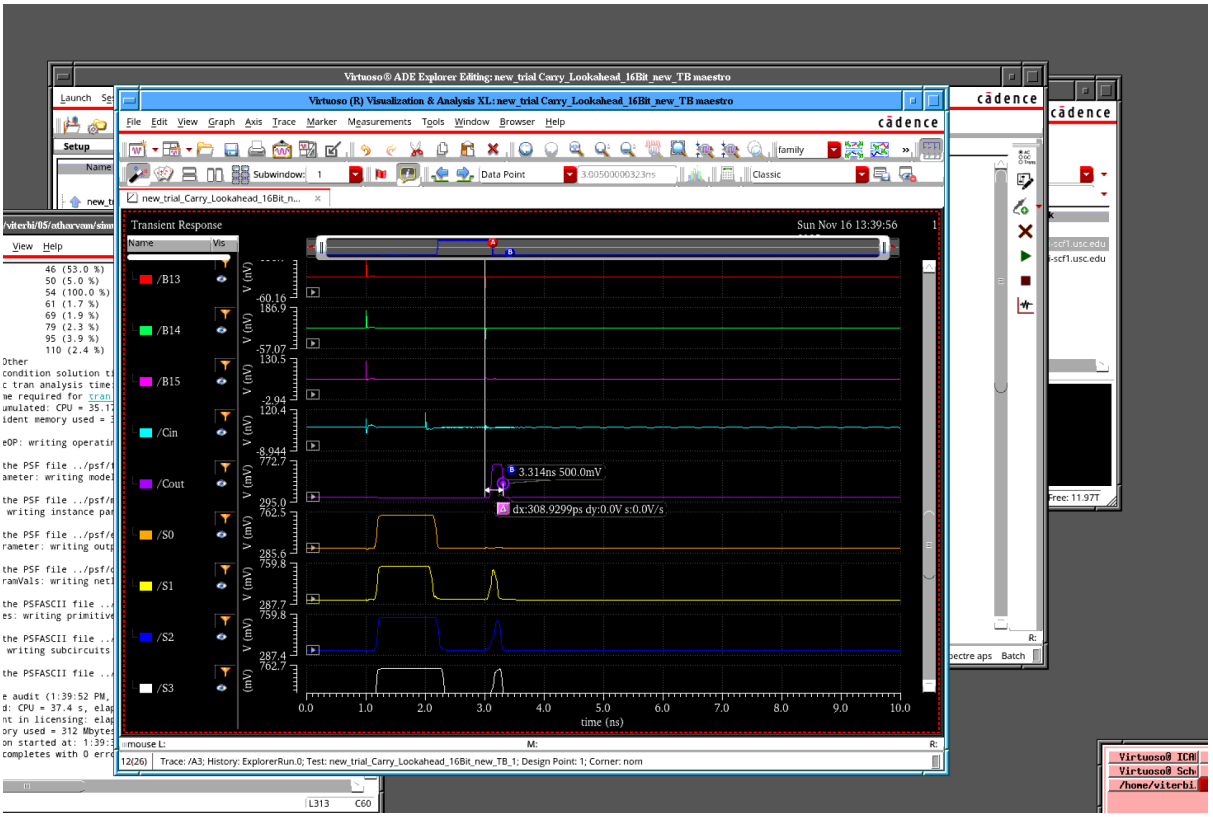
1 NOR gate



Test case for critical path

A = FFFC → FFFD
B = 0003 → 0003
C = 0 → 0



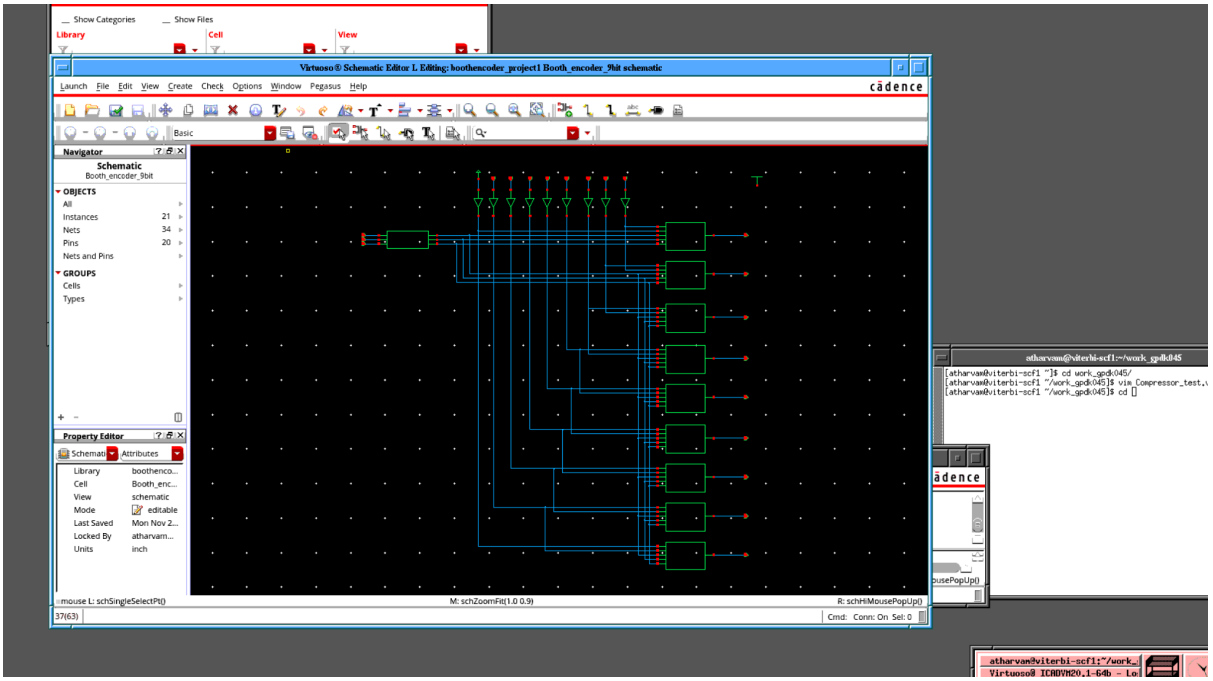


The system has to work at 0.83 Ghz.

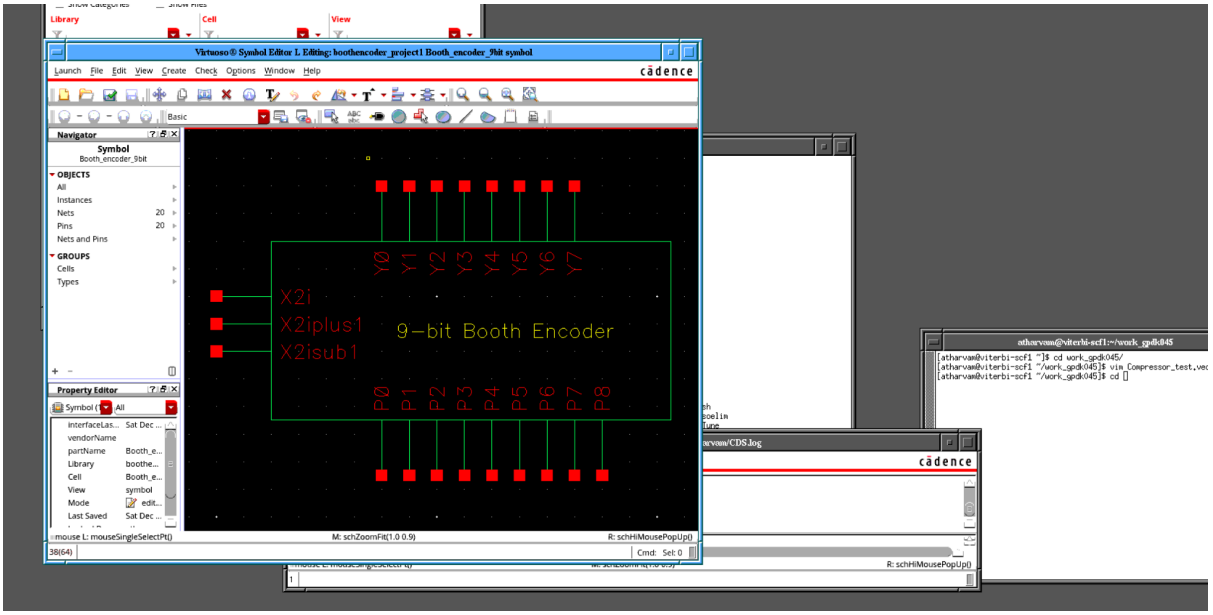
PHASE - II

Part A: Booth Encoder

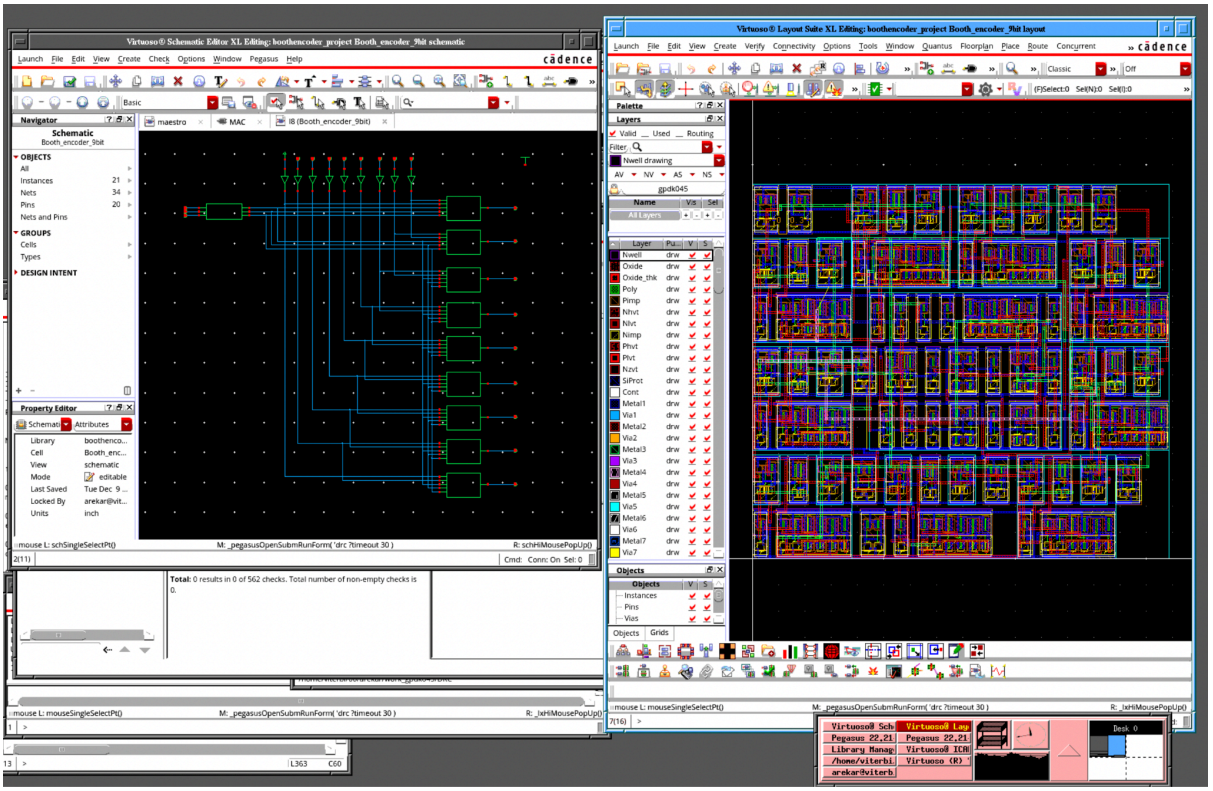
Schematic



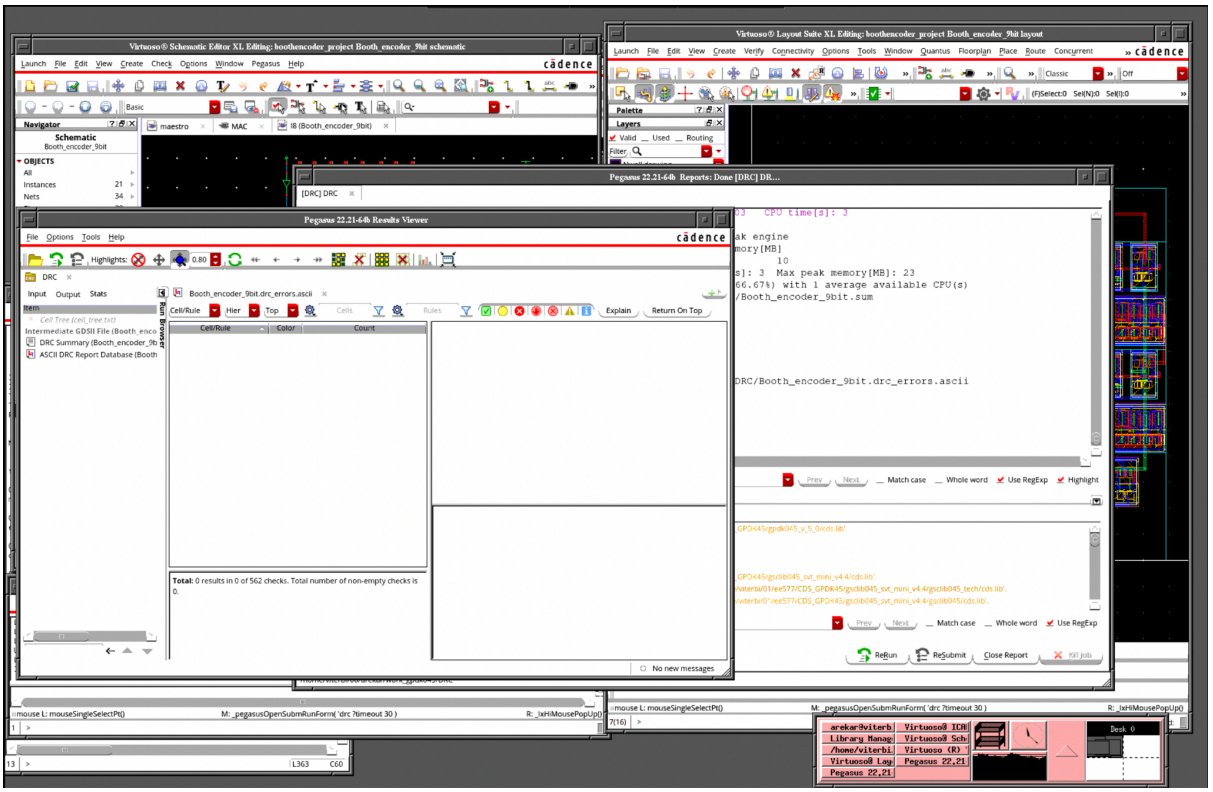
Symbol



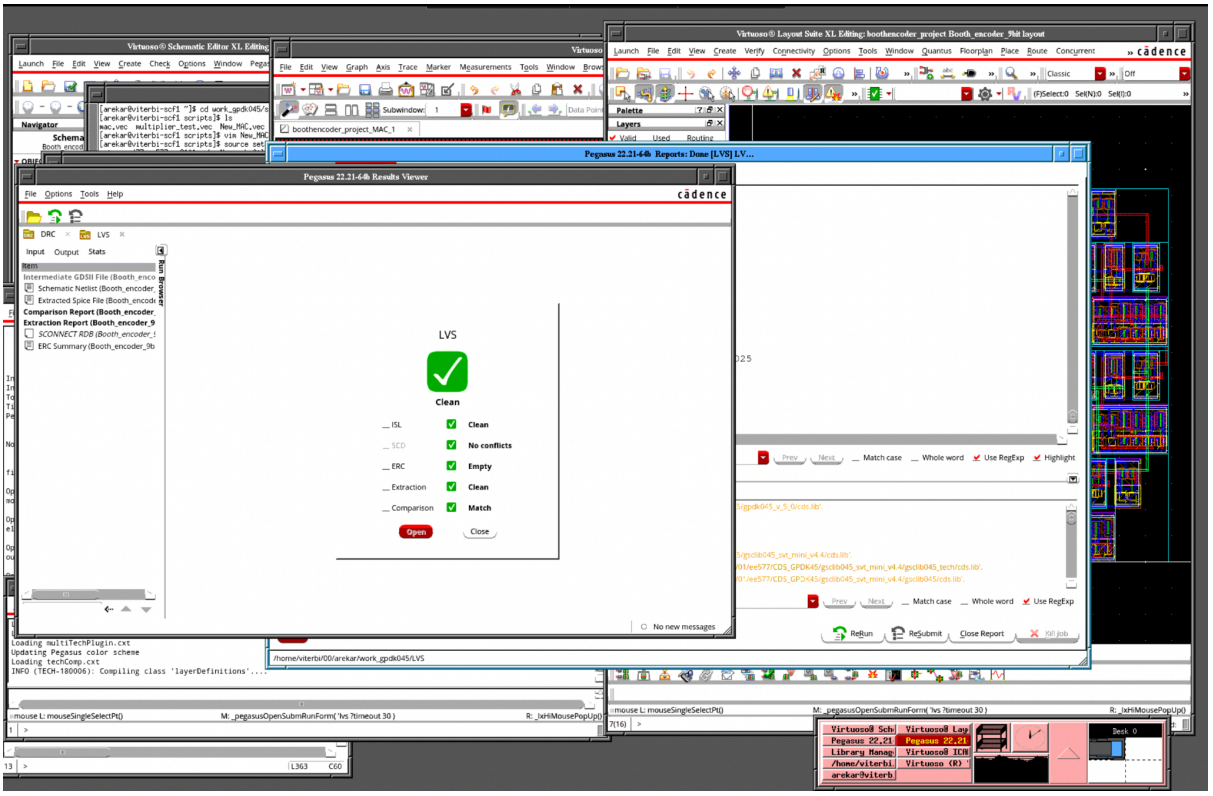
Layout



DRC Report

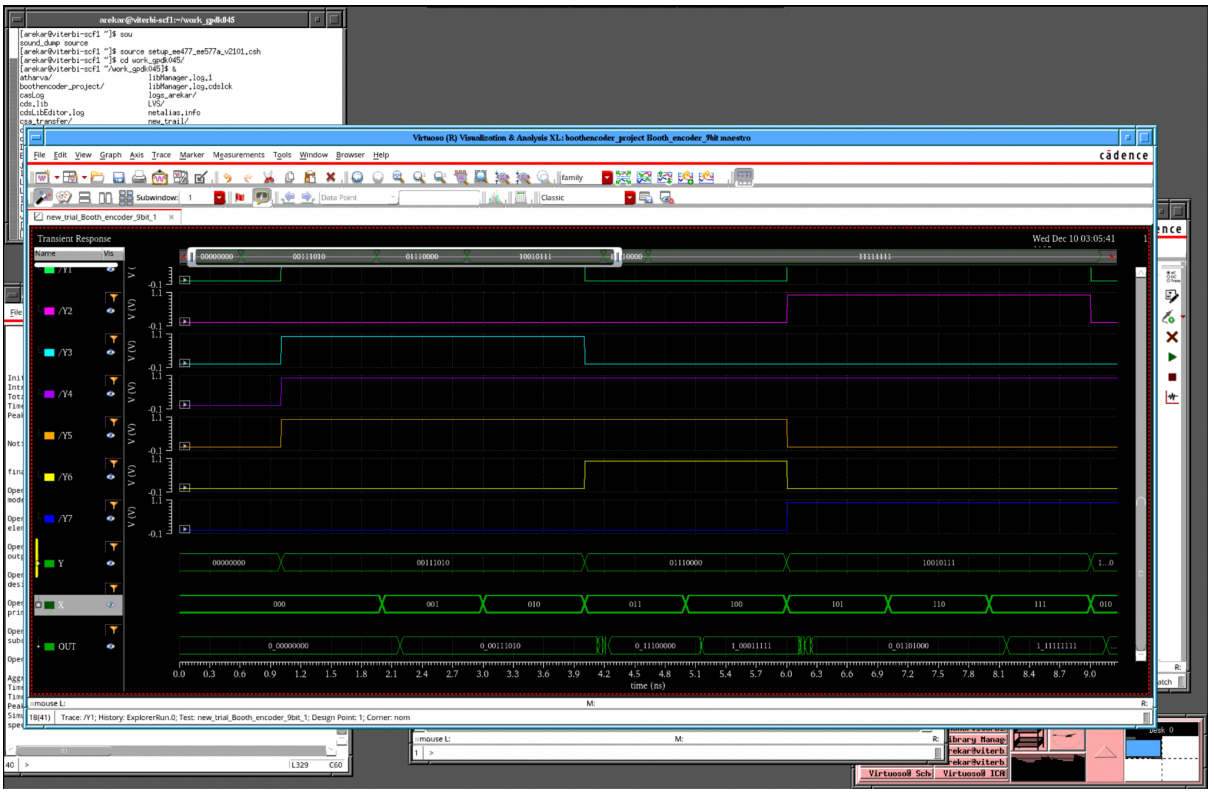


LVS Report

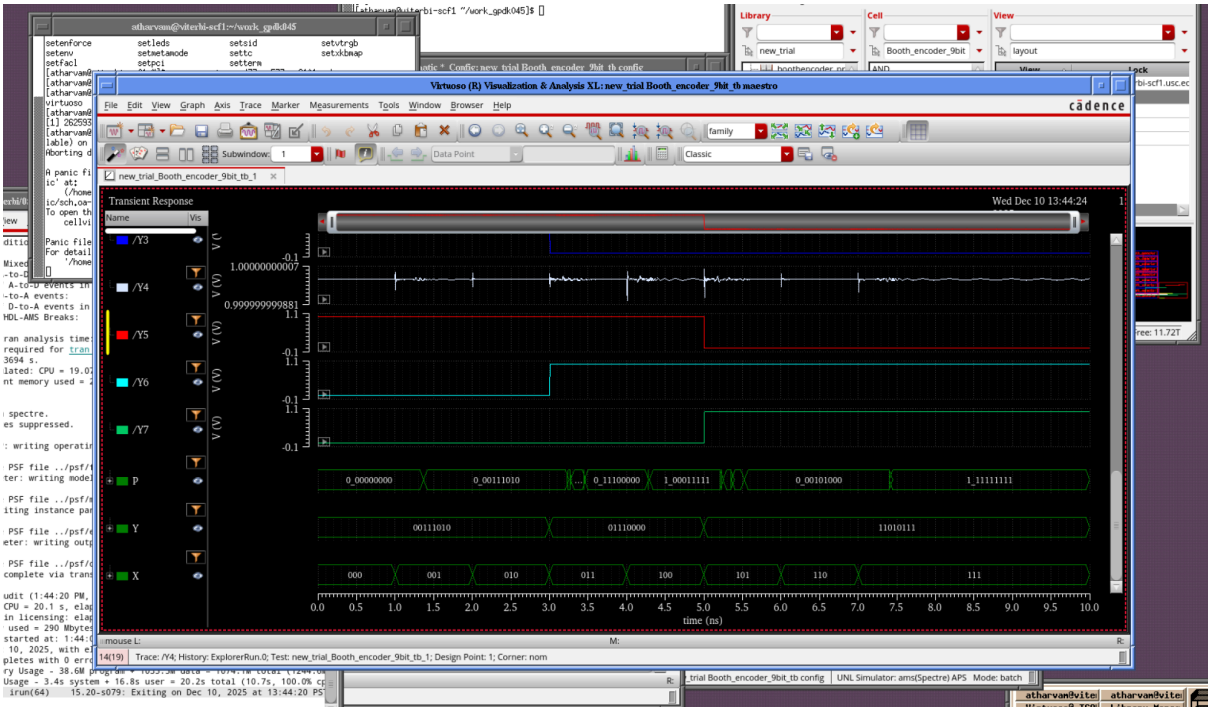


Functionality Verification

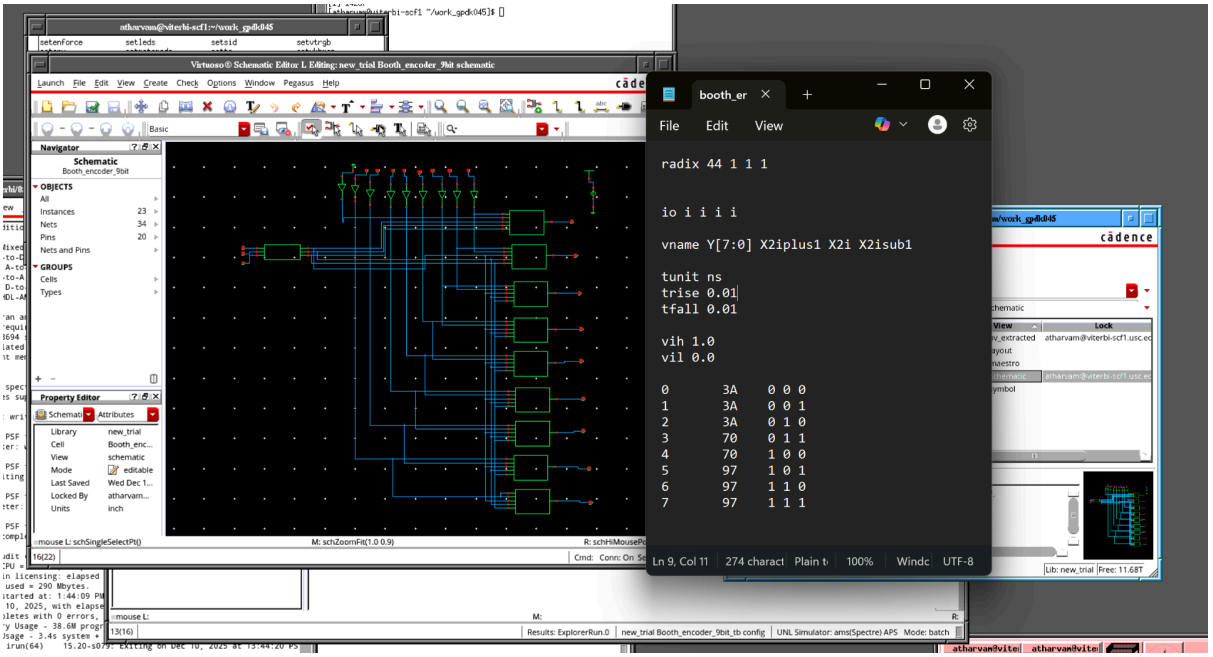
A. Schematic Testing



B. Post-Layout Simulation

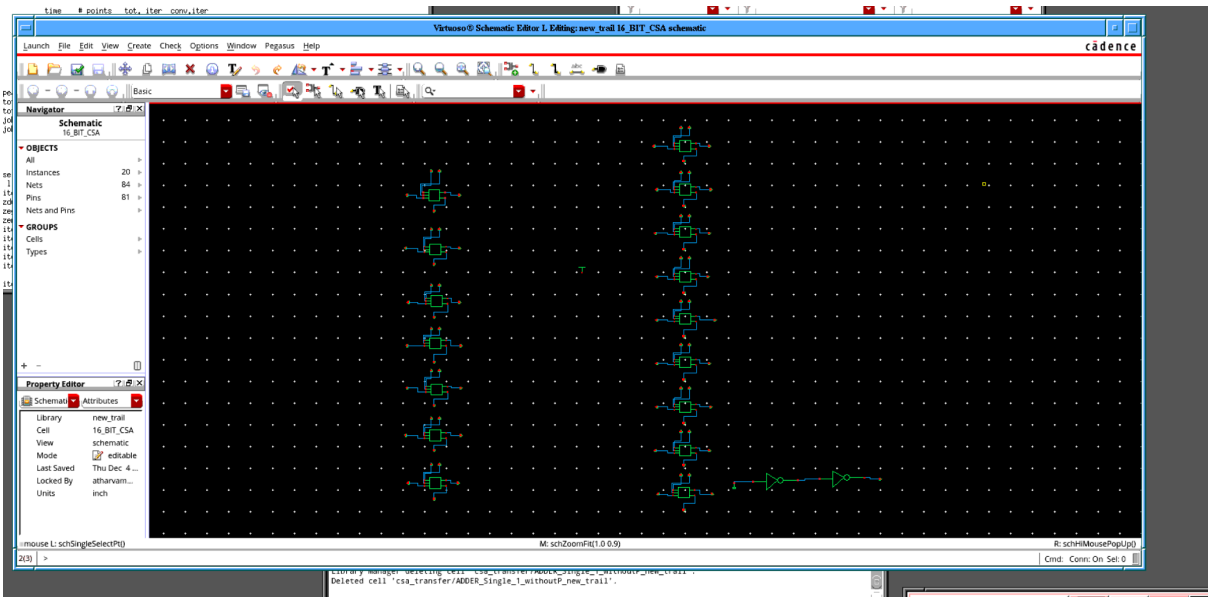


Vector File used for Functionality Verification

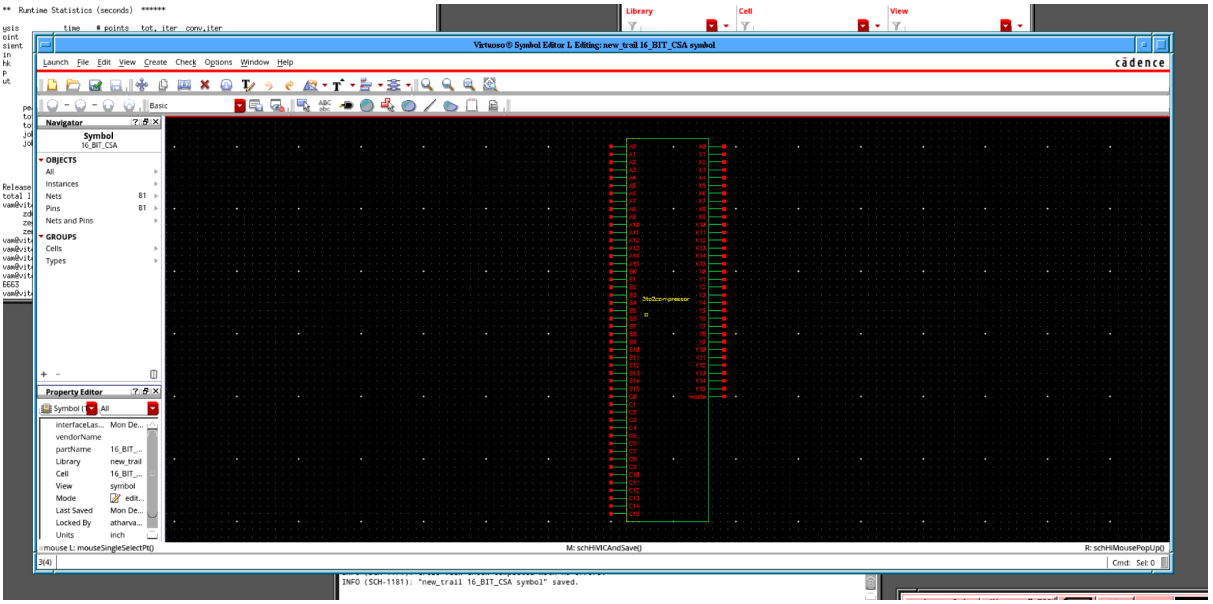


Part B: 16-bit 3-to-2 Encoder

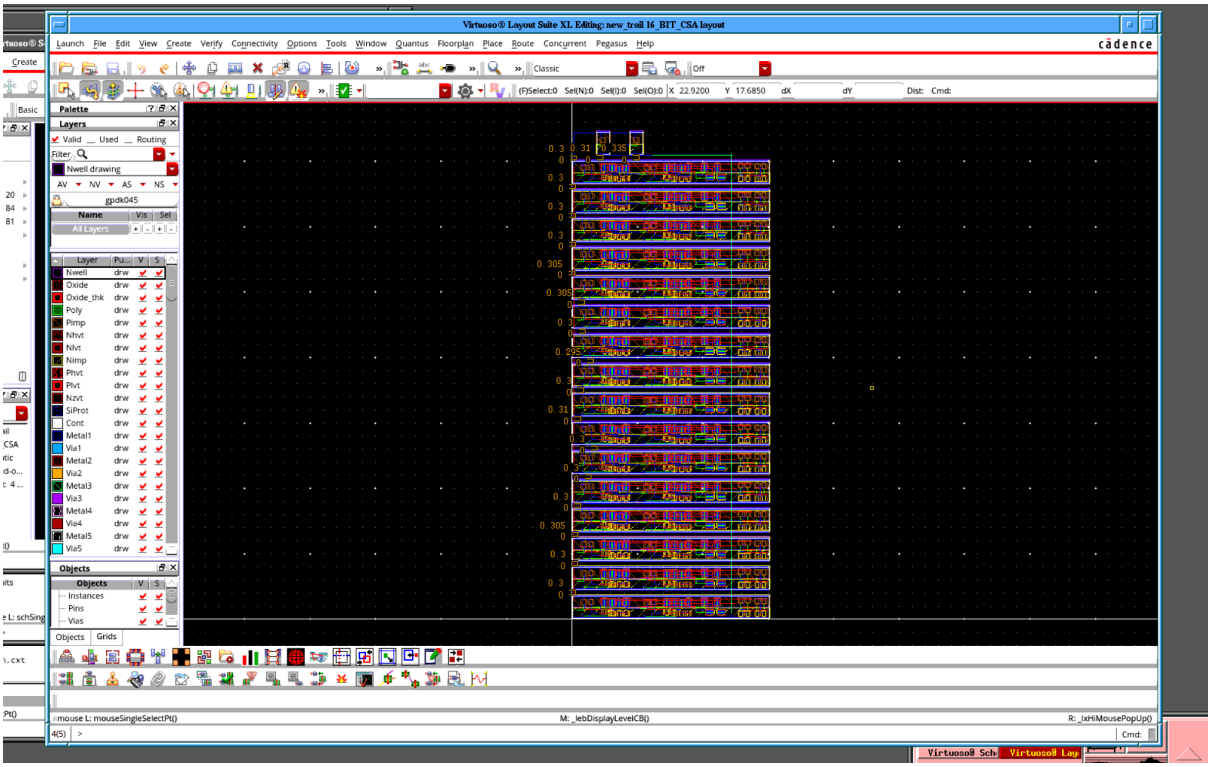
Schematic



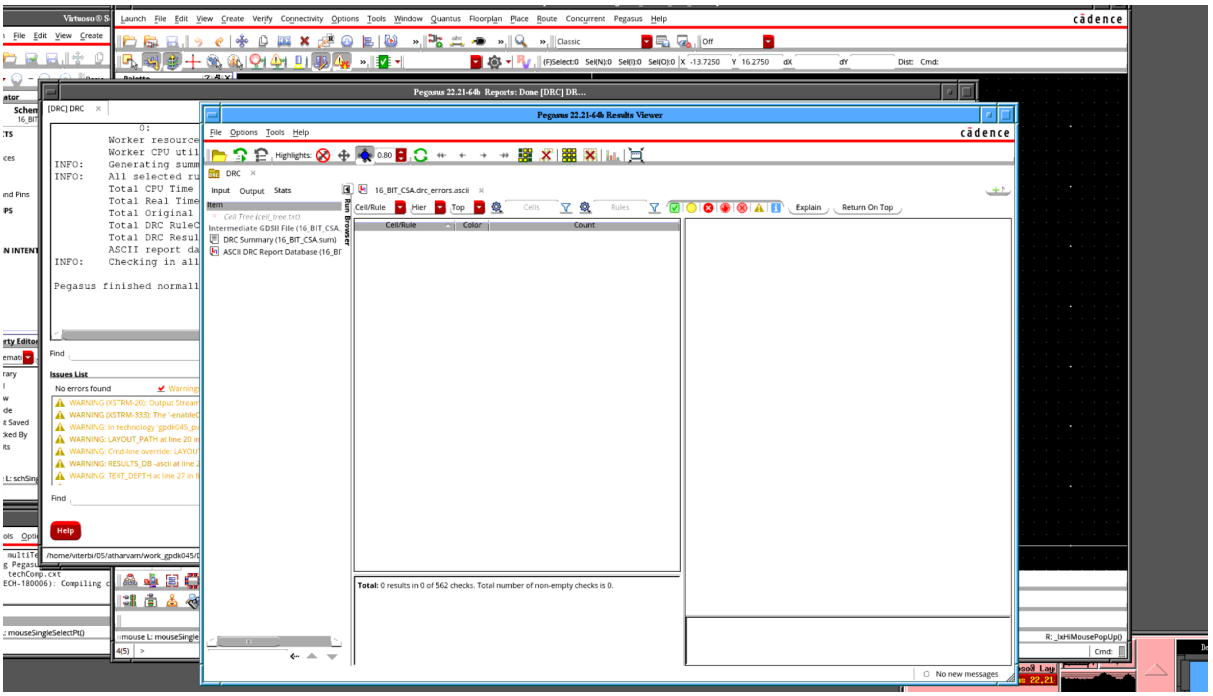
Symbol



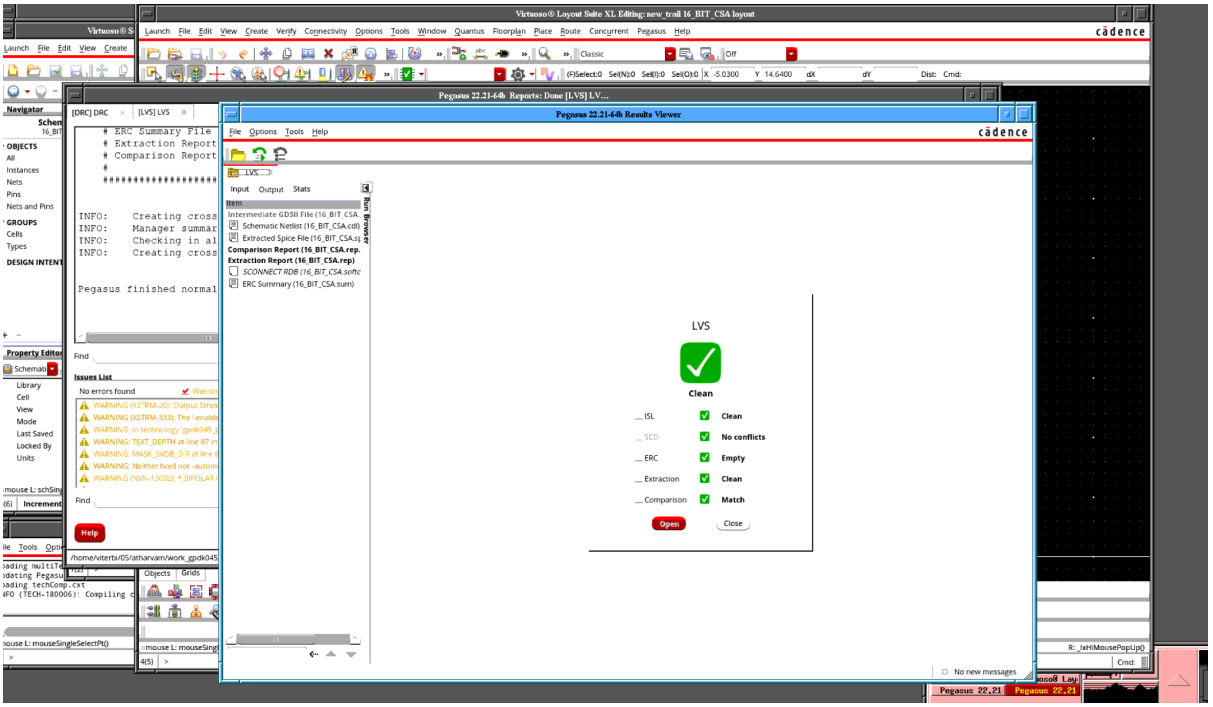
Layout



DRC Report



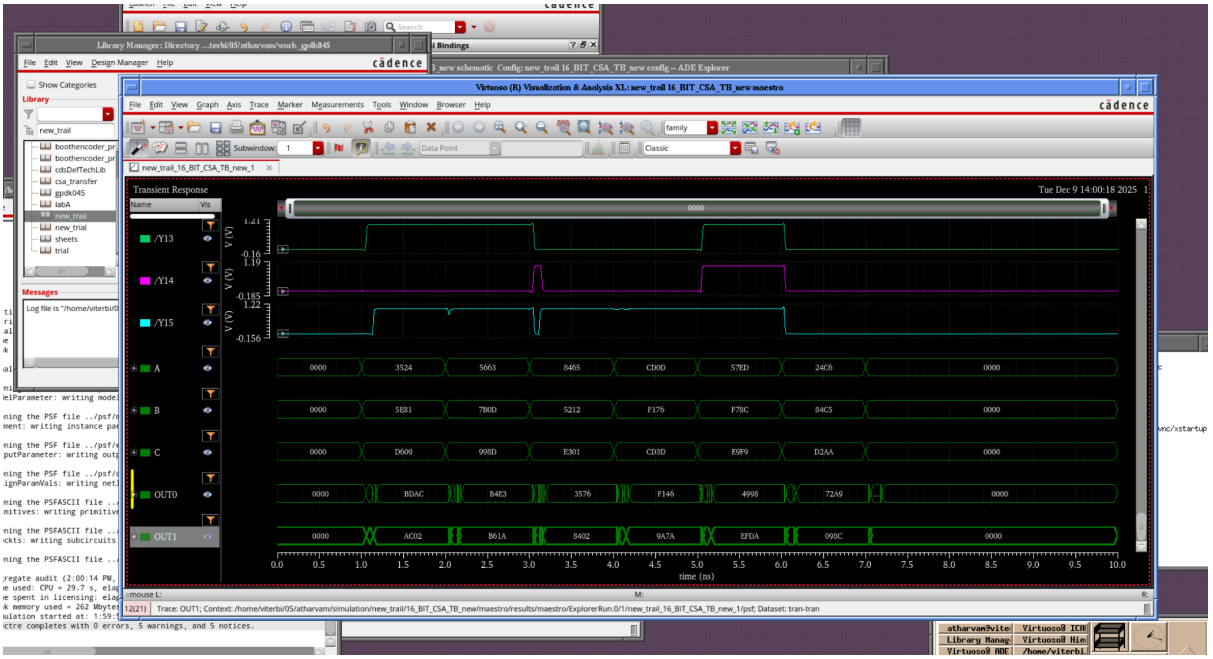
LVS Report



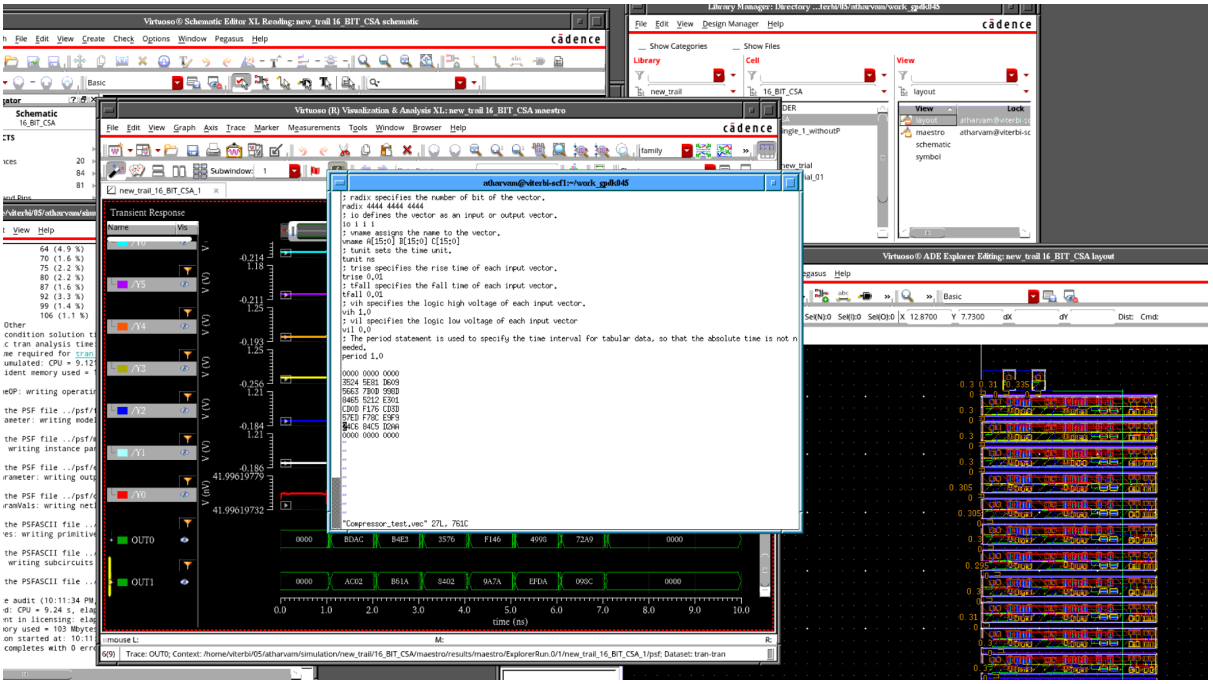
Functionality Verification

A. Schematic Testing

B. Post-Layout Simulation

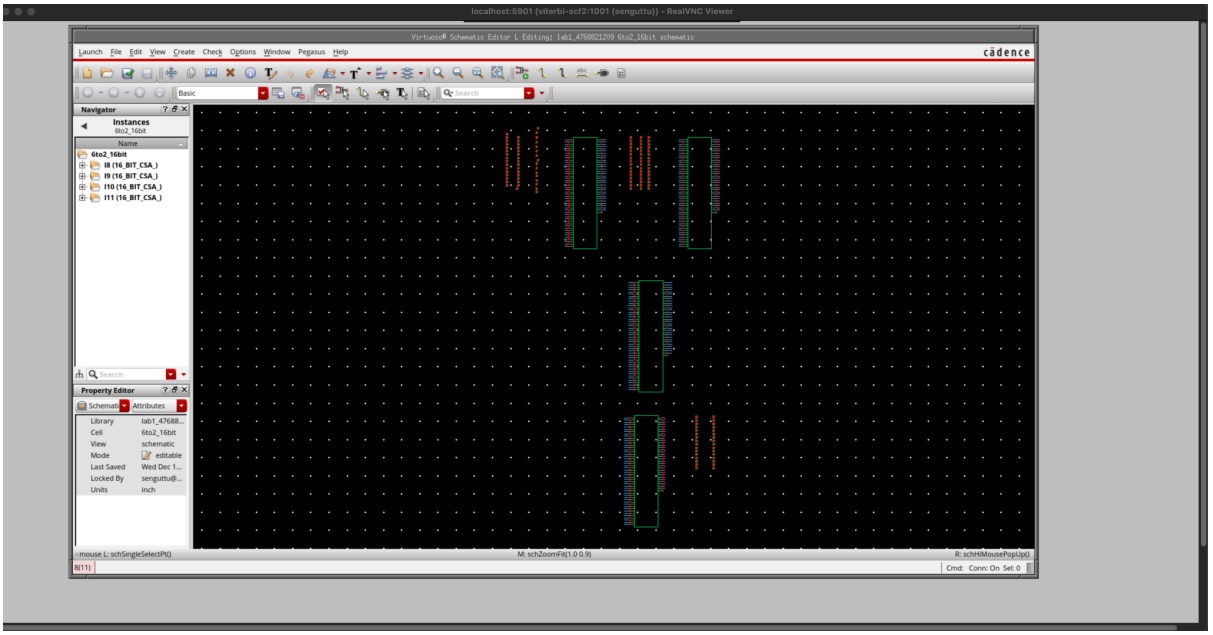


Vector File used for Functionality Verification

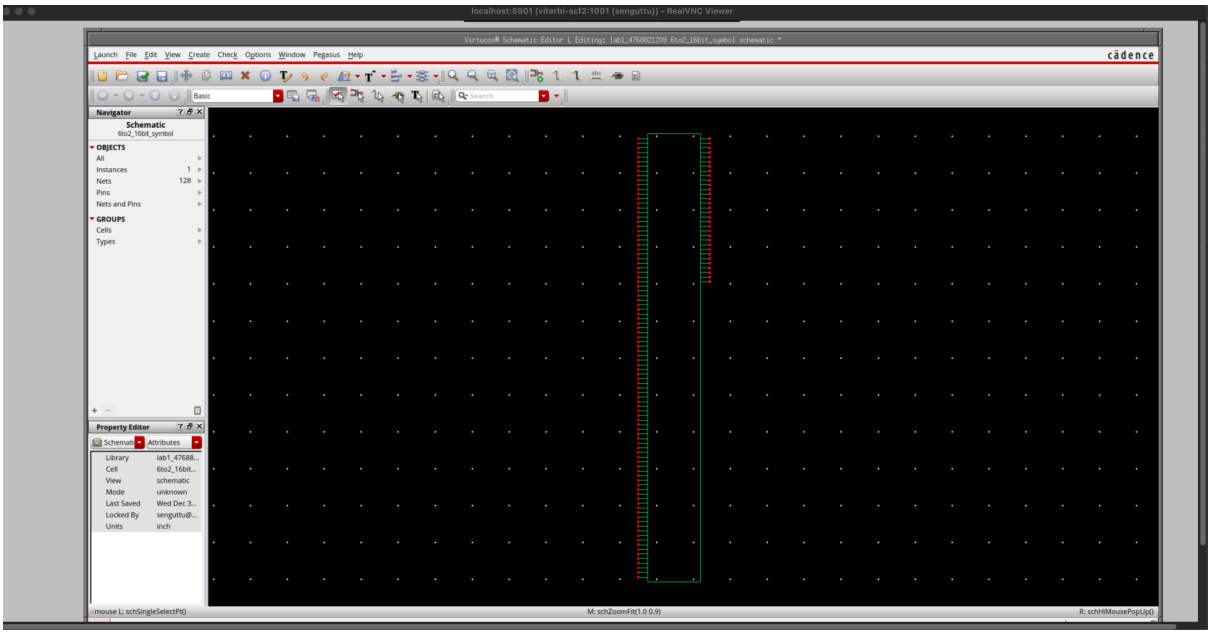


Part C: 16-bit 6-to-2 Encoder

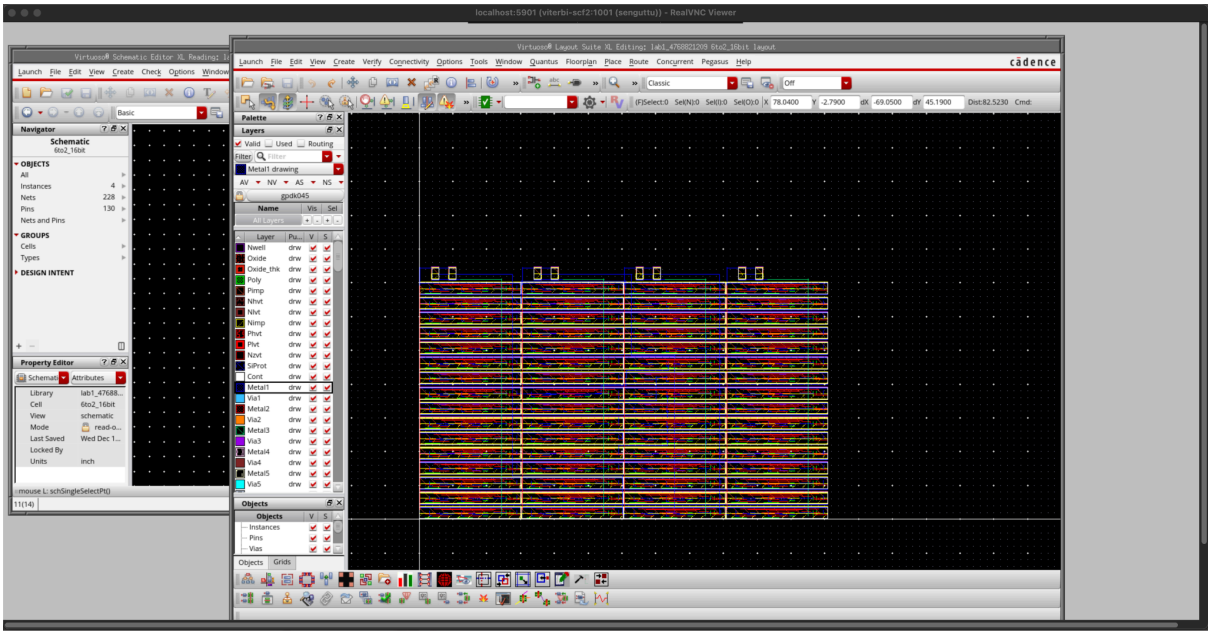
Schematic



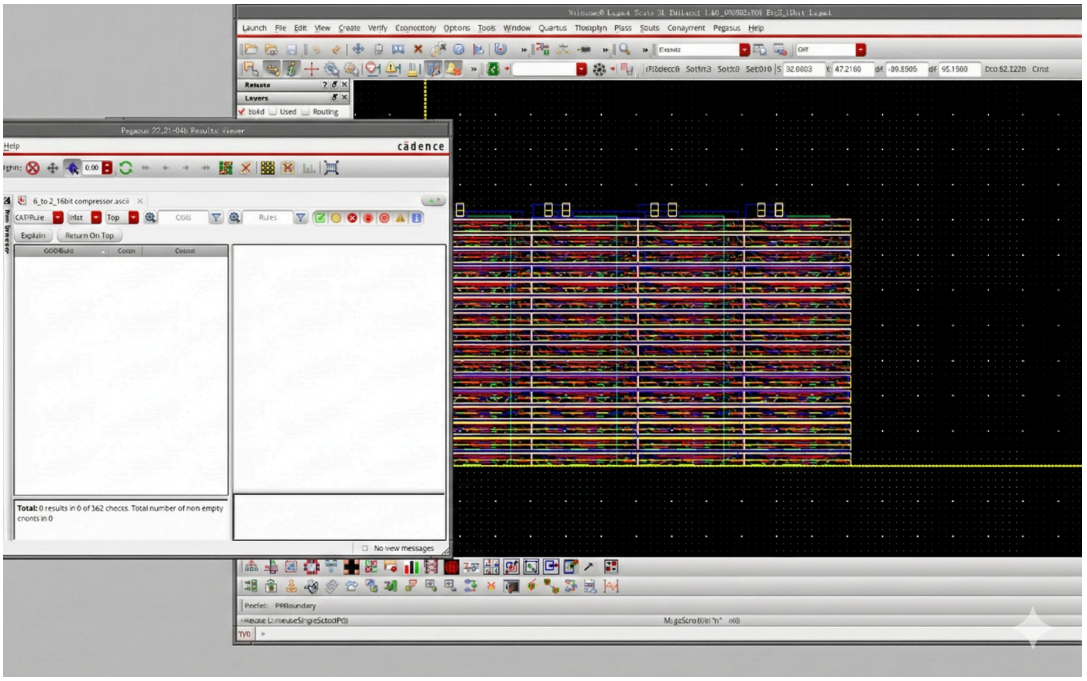
Symbol



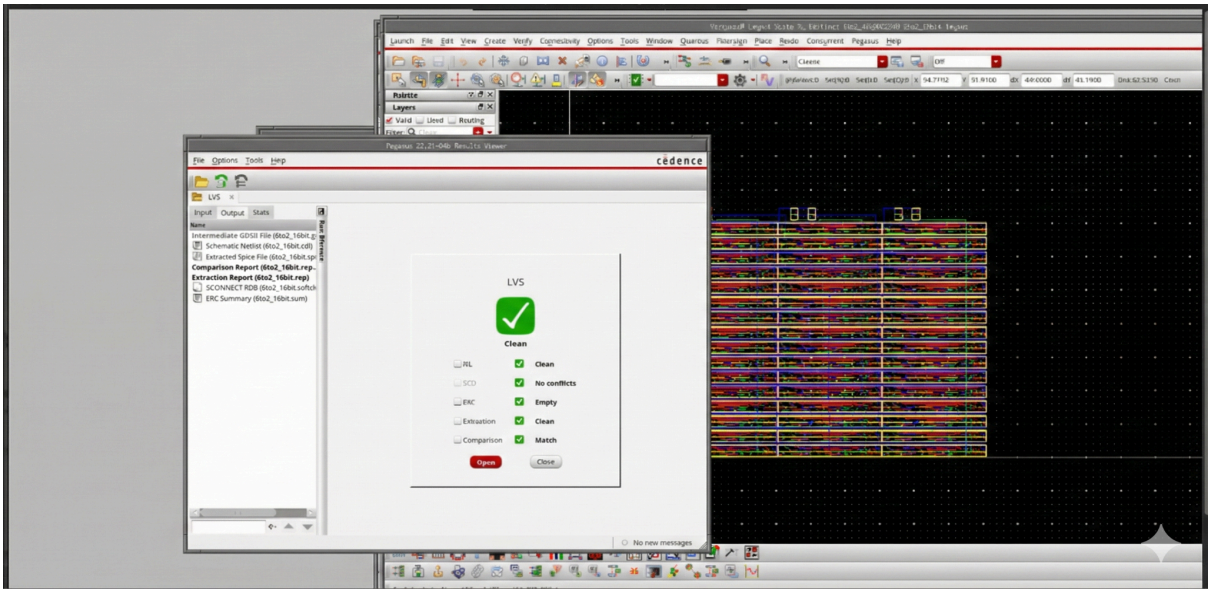
Layout



DRC Report



LVS Report



Functionality Verification

A. Schematic Testing

B. Post-Layout Simulation

Vector File used for Functionality Verification